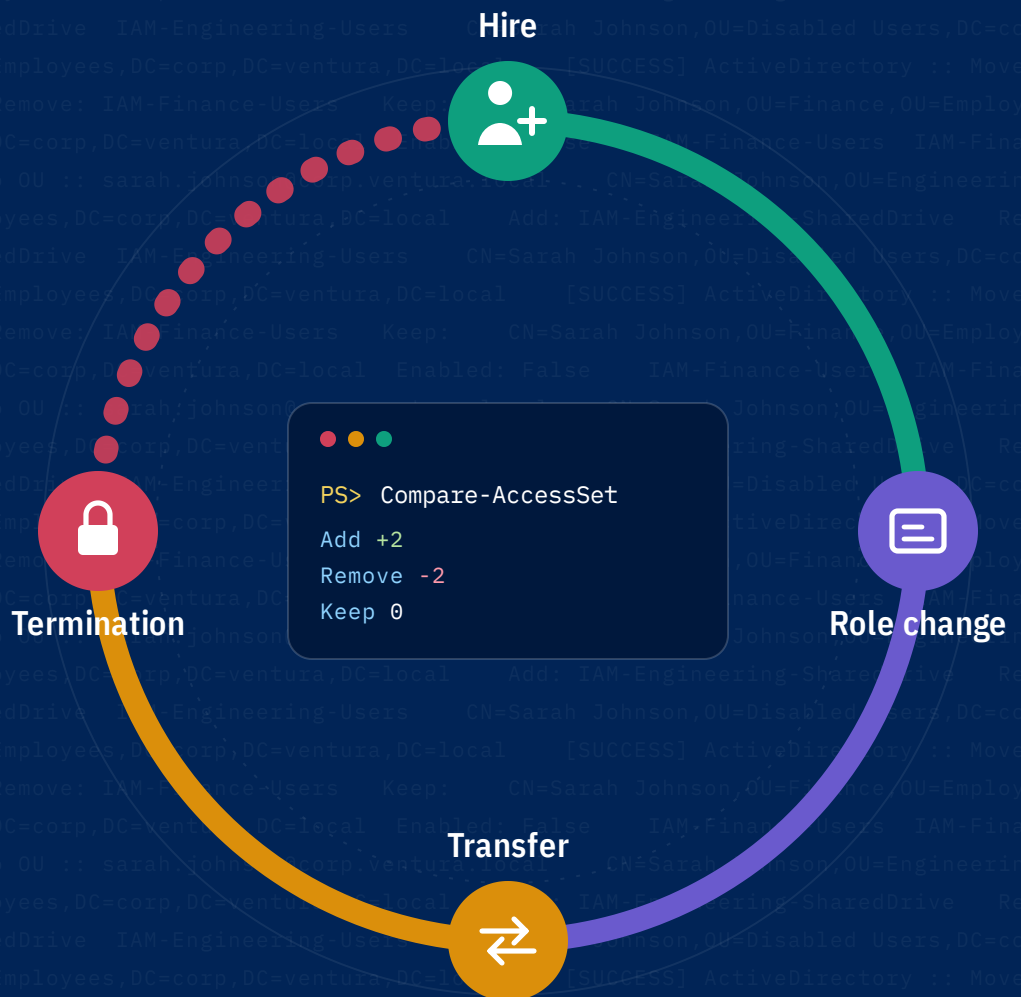


Employee Identity Lifecycle Automation

A PowerShell engine that turns HR events into correct, audited Active Directory access for every hire, role change, transfer and termination.



00 Executive summary

Every organization repeats the same identity work: a person is hired, changes roles, moves departments and eventually leaves. In most IT departments each of those events is a ticket and a checklist that someone works through by hand. Every manual step is a chance to delay a new hire, grant the wrong access, or leave a former employee's account active.

This project replaces that checklist with an engine. HR data, exported from a Workday-style system, is treated as the single source of truth. The PowerShell engine translates each employee's department and job title into the exact access they should hold. It compares that desired state with what the directory actually contains, then adds or removes only the difference, writing every action to an audit log.

I designed and built the full stack in a private Hyper-V lab. It includes a Windows Server 2025 domain controller, a new Active Directory forest, an OU and group access model, a 983-line PowerShell engine, and a scripted test campaign. A single test employee was taken through all four lifecycle events, and every result was verified independently in Active Directory.

4/4 Lifecycle scenarios passed: hire, role change, transfer, termination	17 Directory changes applied by 4 commands, with no manual console edits	30 Purpose-built PowerShell functions across 7 functional layers	4 Target platforms modeled: AD (validated live), Entra ID, Intune, AVD
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What the engine guarantees

Access follows HR data, not memory. Moves remove old access automatically, so privileges never pile up. Leavers are disabled, stripped of managed groups and quarantined in one pass. Every run can be previewed with `-WhatIf`, re-run safely, and traced in a JSONL audit trail.

Honest scope

The Active Directory path was executed and verified end to end. The Microsoft Entra ID, Intune and Azure Virtual Desktop paths are fully implemented in code but were run with `-SkipCloud`, because the lab has no cloud tenant yet. Connecting one is the next milestone in the roadmap.

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A note on redaction. All screenshots are real captures from the lab. Secrets and network identifiers were removed before publication and are marked with **REDACTED** labels. Internal addresses appear as **10.x.x.171**. Screenshots are shown uncropped, two per page at near full page width so every detail stays readable. Window titles identify the lab VM **Esmeralda_Ventura_DC01** on host **ESME_VENTURA**.

01 The problem: lifecycle access is still a manual checklist

Identity teams call it Joiner-Mover-Leaver (JML). It is one of the highest-volume and highest-risk workflows in IT, because access is the front door to every system a company owns.

A typical manual process starts with an email or ticket from HR. A technician opens Active Directory Users and Computers, types the user's details, picks an OU, and often copies group memberships from "someone in the same team." Cloud apps, device policies and virtual desktops follow in other consoles. Each hand-off relies on someone reading the ticket correctly and remembering every step.

Where it breaks

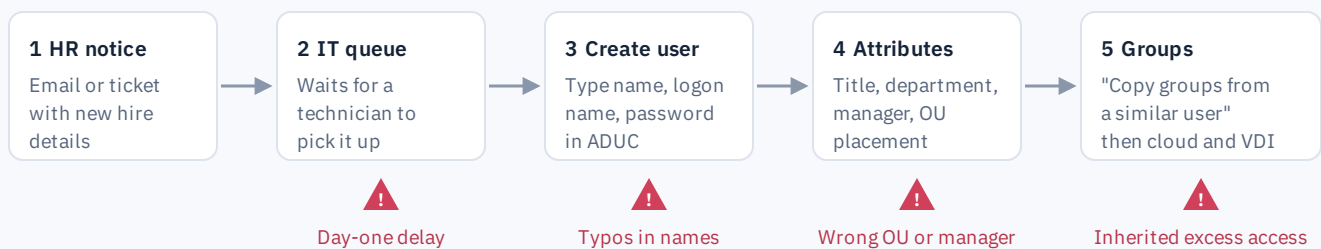
Joiners wait for access, or inherit someone else's permissions through "copy user."

Movers receive new access, but old access is rarely removed, so privileges accumulate over years.

Leavers depend on a ticket reaching IT on time. A missed step leaves an active account with a former employee.

A manual joiner, step by step

Every box is a person doing work by hand. Every red marker is a common failure point.



What happens later

Mover: access only ever grows

Finance groups + Engineering groups = Both, forever

New access is requested loudly. Removal of old access has no requester, so it is quietly skipped. This is privilege creep, and auditors look for it first.

Leaver: the account outlives the job

Last day ----> Ticket late --> Still enabled

An enabled account with group memberships remains a working credential. It becomes an orphaned account and a security and compliance finding.

Figure 1. The manual JML process and its typical failure points. The automation in this report removes the human hand-off from every step shown.

Design goal

Make HR data the only input a person has to get right. Everything downstream, from account creation to group removal, should be computed, applied consistently, safe to preview, and provable after the fact.

02 Solution architecture

The engine follows a desired-state model, the same idea behind infrastructure-as-code tools. Instead of scripting individual clicks, it declares what access should exist and reconciles reality to match.



Figure 2. End-to-end architecture. Solid arrows show paths executed in the lab. Dashed arrows show cloud connectors that are implemented and gated behind -SkipCloud.

02 Engineering principles built into the code

Identity automation can do a lot of damage very quickly. These guardrails are what make the engine safe to run against a production directory.

- **HR is the source of truth**

Access is derived only from department, job title and status in the HR record. Users are matched on the immutable `employeeID` attribute, never on display name.
- **Desired state, not scripted clicks**

`Compare-AccessSet` returns Add, Remove and Keep lists. The engine applies only the difference, so a transfer removes old access as naturally as it grants new access.
- **Idempotent by design**

Re-running the same input makes no duplicate changes. The role-change test proved this: groups already correct were kept, with no redundant writes.
- **Preview before you apply**

`SupportsShouldProcess` with `ConfirmImpact = 'High'` enables native `-WhatIf` and confirmation prompts. `-AuditOnly` logs the full plan as "Planned" without touching anything.
- **Limited blast radius**

The engine only manages groups whose names start with `IAM-`. Built-in and hand-managed groups such as Domain Users are never modified, as every test confirmed.
- **Every write is audited**

All changes pass through one function, `Invoke-ManagedChange`, which records Success, Skipped or Failed with the error message. There is no way to change access off the record.
- **Fail safe, fail loudly**

`Set-StrictMode` and `ErrorActionPreference = 'Stop'` turn silent errors into exceptions. Each employee runs in its own try/catch, so one bad record is logged without halting the batch.
- **Leavers are quarantined, not deleted**

Terminated accounts are disabled, removed from managed groups and moved to a Disabled Users OU. This preserves the SID and history for audits, legal holds and rehire.

One command per lifecycle event

```
# Dry run: print the reconciliation plan and simulate every change
.\Invoke-EmployeeIdentityLifecycle.ps1 -WorkdayCsv .\EmployeeLifecycleTestCases\Workday-Hire.csv `
  -ConfigPath .\AccessMappings.json -SkipCloud -WhatIf

# Real execution: same command without -WhatIf
.\Invoke-EmployeeIdentityLifecycle.ps1 -WorkdayCsv .\EmployeeLifecycleTestCases\Workday-Hire.csv `
  -ConfigPath .\AccessMappings.json -SkipCloud
```

The same entry point handles all four events. The event type comes from the `LifecycleEvent` column in the HR record, not from the operator.

03 Lab environment build

To test identity automation honestly, you need a real directory. I built one from bare metal: a Hyper-V virtual machine promoted to the first domain controller of a brand-new Active Directory forest.

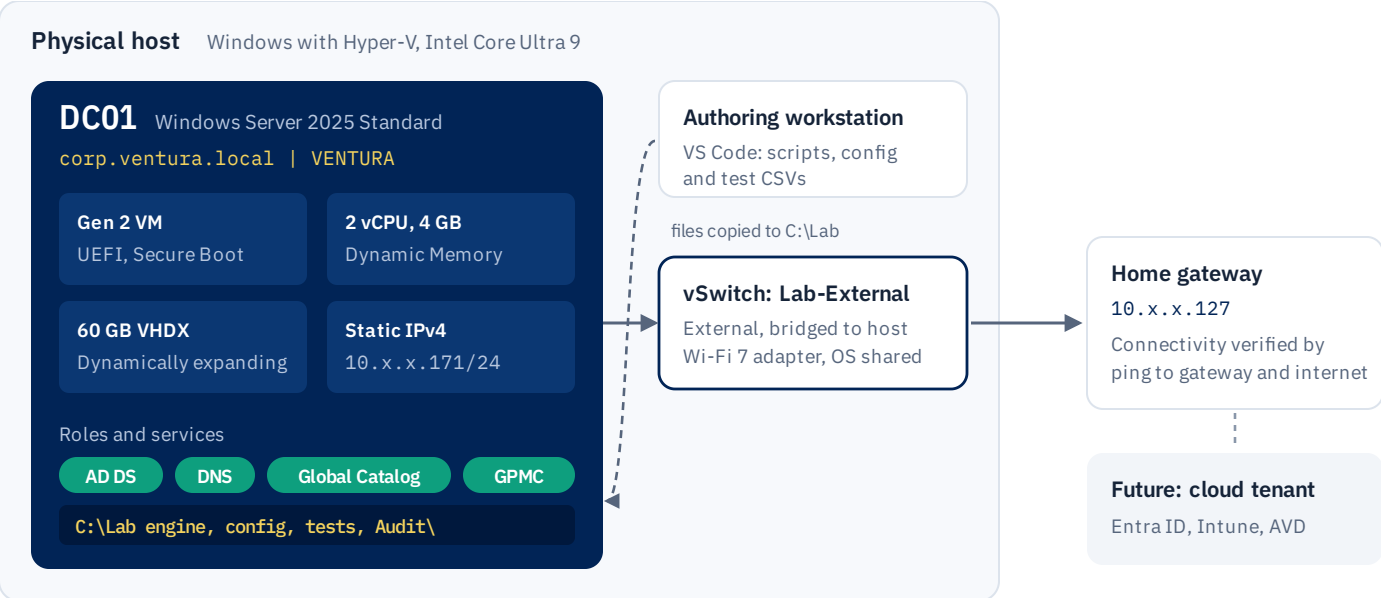
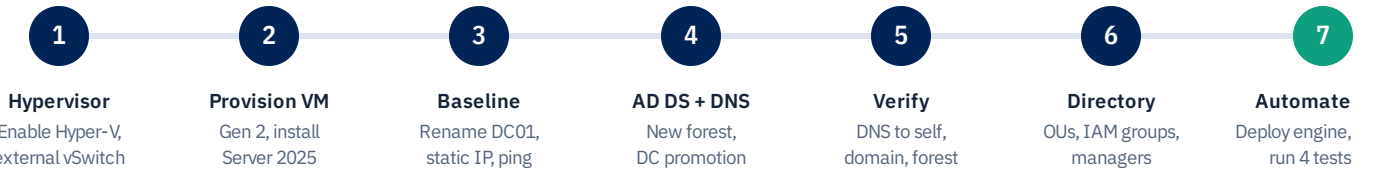


Figure 3. Lab topology. Addresses are masked. The domain controller runs its own DNS and is its own global catalog.

Build sequence



Steps 1 to 6 are documented from page 7. Step 7, the automation itself, is covered from page 20 onward.

1 DC Single-site forest: DC01 runs AD DS, DNS and the global catalog	2025 Forest and domain functional level, the newest available	3 + 1 Department OUs, plus a Disabled Users quarantine OU	6 Managed IAM- security groups, 2 manager accounts
--	---	---	--

03 Build log: steps 1–3

Steps 1–2: Hypervisor and virtual machine

Hyper-V was enabled on the physical host `ESME_VENTURA`, and a virtual machine named `Esmeralda_Ventura_DC01` was built to become the domain controller.

Setting	Choice	Why it matters
Network	External vSwitch <code>Lab-External</code>	Puts DC01 on a real network with internet access for activation and updates, like a branch server.
Generation	Generation 2, Secure Boot on	UEFI firmware and Secure Boot match modern server hardening baselines.
Compute	2 vCPU, 4,096 MB, Dynamic Memory (512 MB minimum)	Enough for AD DS and DNS while leaving host resources for other lab VMs.
Storage	60 GB dynamically expanding <code>.vhdx</code>	Room for NTDS, SYSVOL and logs, consuming disk only as it grows.
OS	Windows Server 2025 Standard Evaluation (Desktop Experience)	Current server release, with the GUI for ADUC, Server Manager and screenshot evidence.

Step 3: Server baseline

Before promotion, the default `WIN-...` hostname was changed to `DC01` (Figure 8), because renaming a domain controller later is far more disruptive. A static IPv4 address, subnet mask, gateway and DNS server were set by hand (Figure 9), since clients locate a domain controller through DNS and its address must never change. Pings to the gateway and to the internet confirmed connectivity.

Troubleshooting a misleading warning

The warning

While adding AD DS and DNS, the wizard reported *"No static IP addresses were found on this computer"*, even though a static IPv4 address had just been set (Figure 10).

The diagnosis

`ipconfig /all` showed **DHCP Enabled: No** and the static IPv4 address marked Preferred (Figure 11). Only IPv6 link-local autoconfiguration remained, a well-known trigger for this generic check.

The decision

With IPv4 proven static and the gateway answering pings, I continued the install. I documented the reasoning so the choice stays deliberate and reviewable.

Uncropped screenshots for these steps follow as Figures 4–11, two per page.

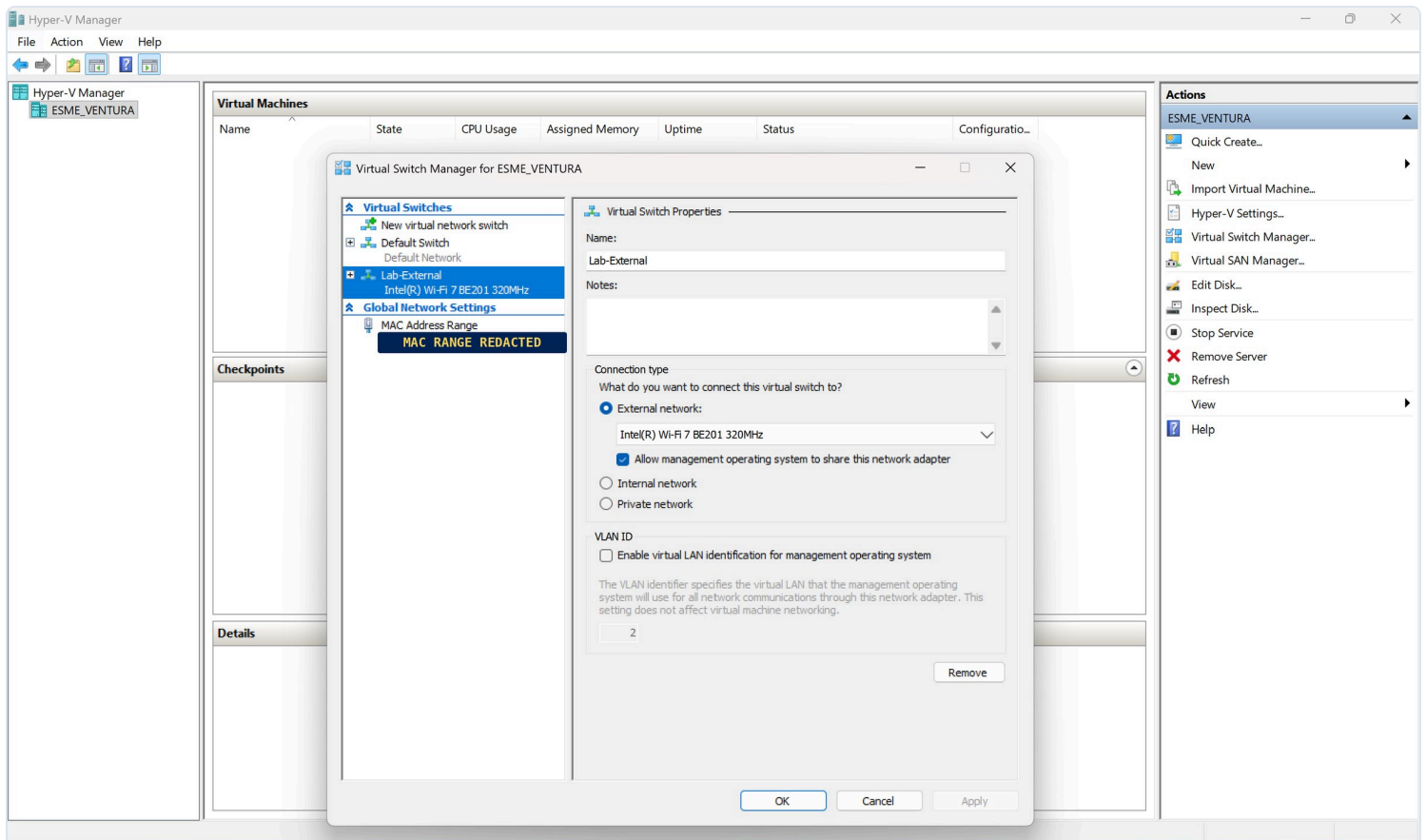


Figure 4. Hyper-V Virtual Switch Manager on host **ESME_VENTURA** : creating **Lab-External** as an external network bound to the host's Wi-Fi 7 adapter, with management OS sharing allowed. The MAC address range is redacted.

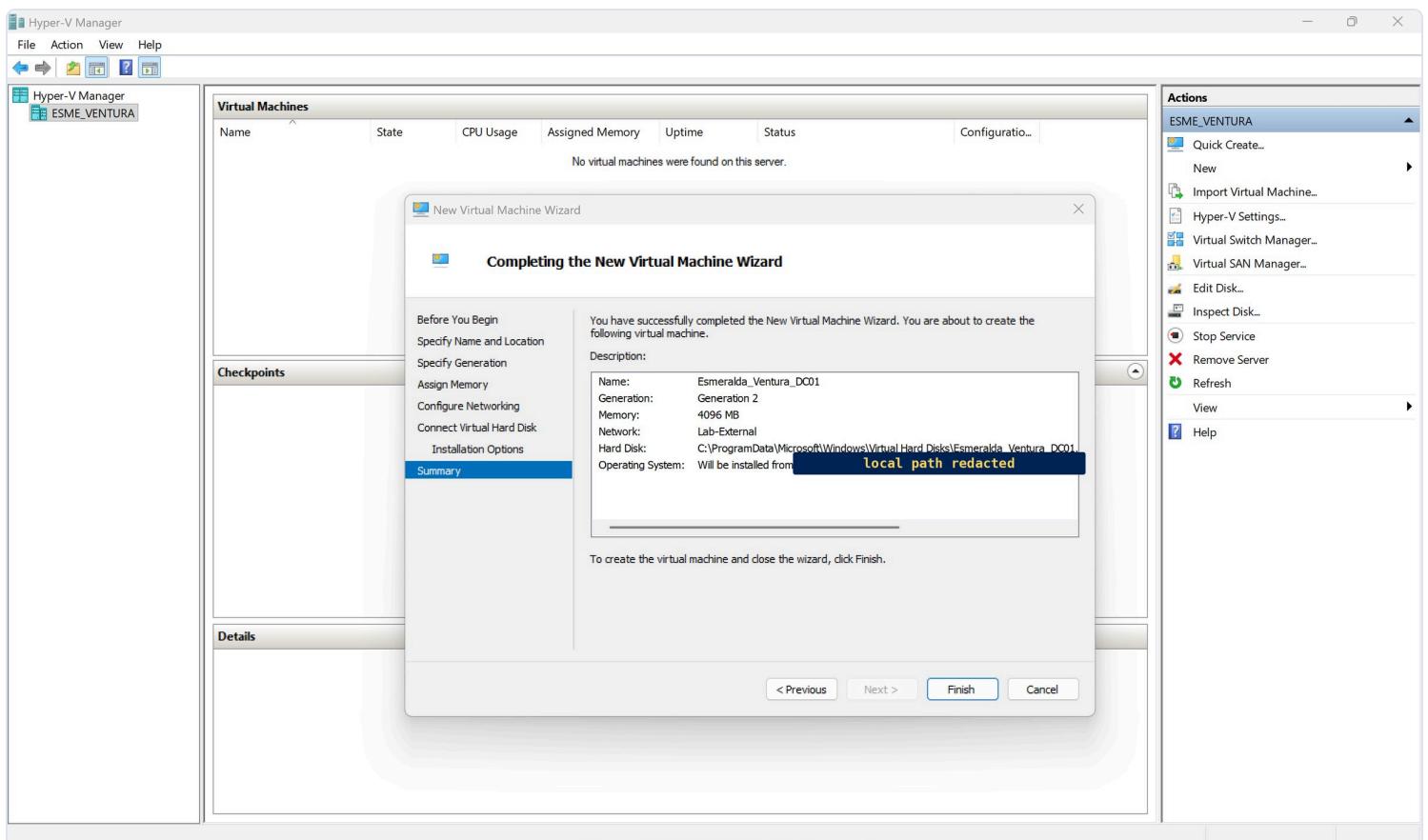


Figure 5. Completing the New Virtual Machine Wizard for **Esmeralda_Ventura_DC01** : Generation 2, 4,096 MB startup memory, the Lab-External network and a new virtual hard disk. The local ISO path is redacted.

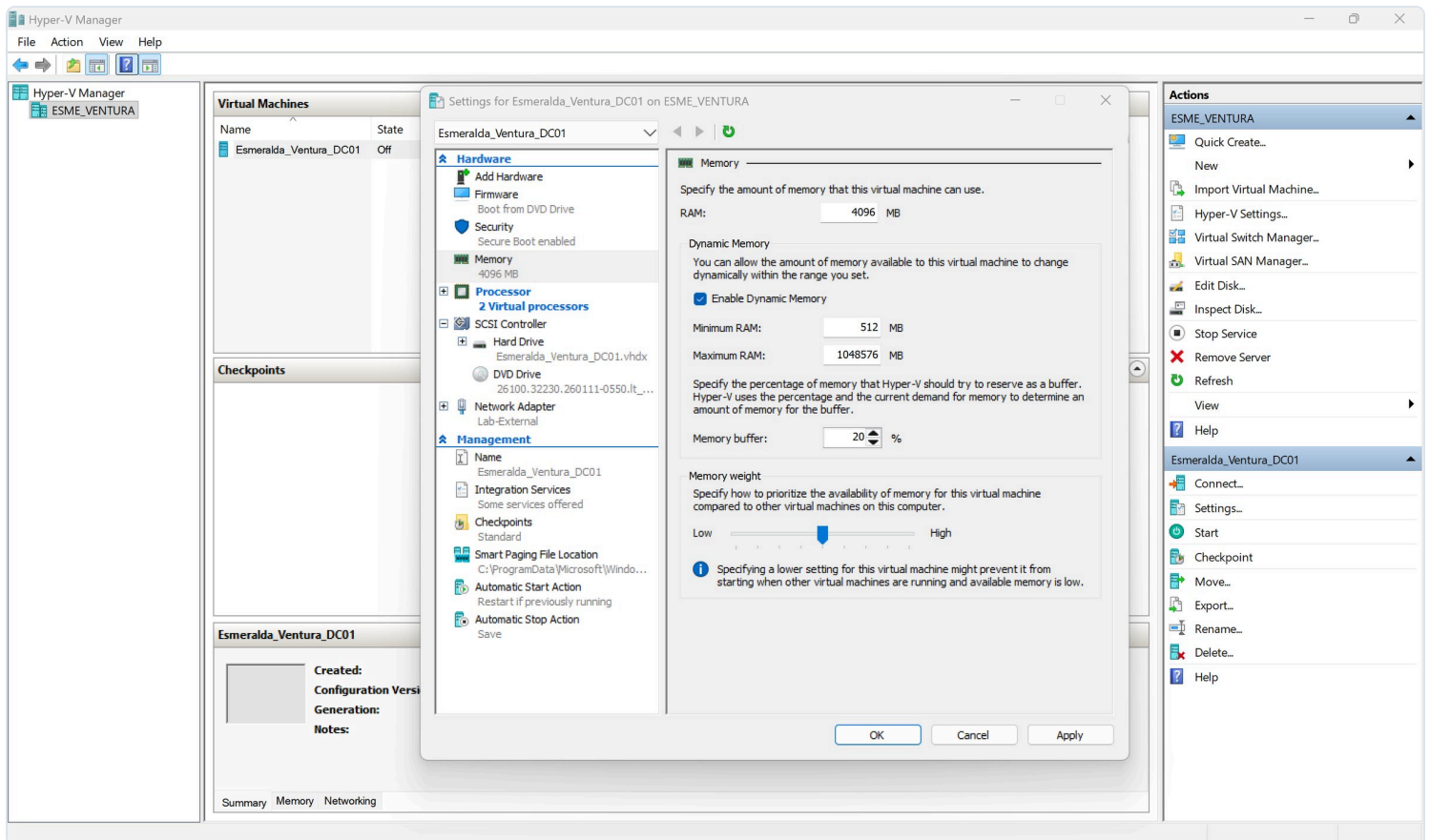


Figure 6. Settings for `Esmeralda_Ventura_DC01` : Dynamic Memory enabled (512 MB minimum, 20% buffer), 2 virtual processors, Secure Boot enabled, and boot from the DVD drive for installation.

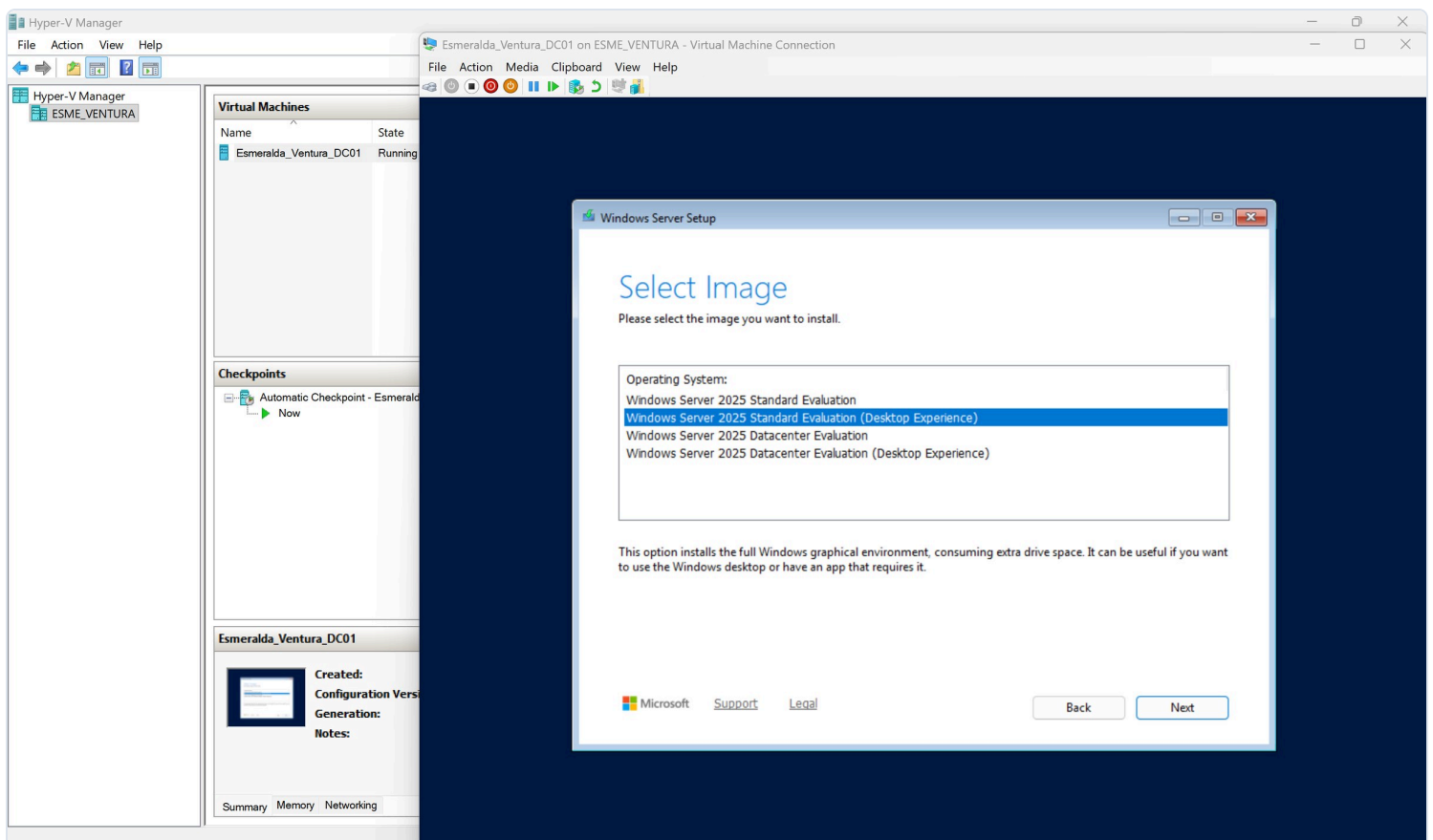


Figure 7. Windows Server Setup running inside the VM: selecting **Windows Server 2025 Standard Evaluation (Desktop Experience)**.

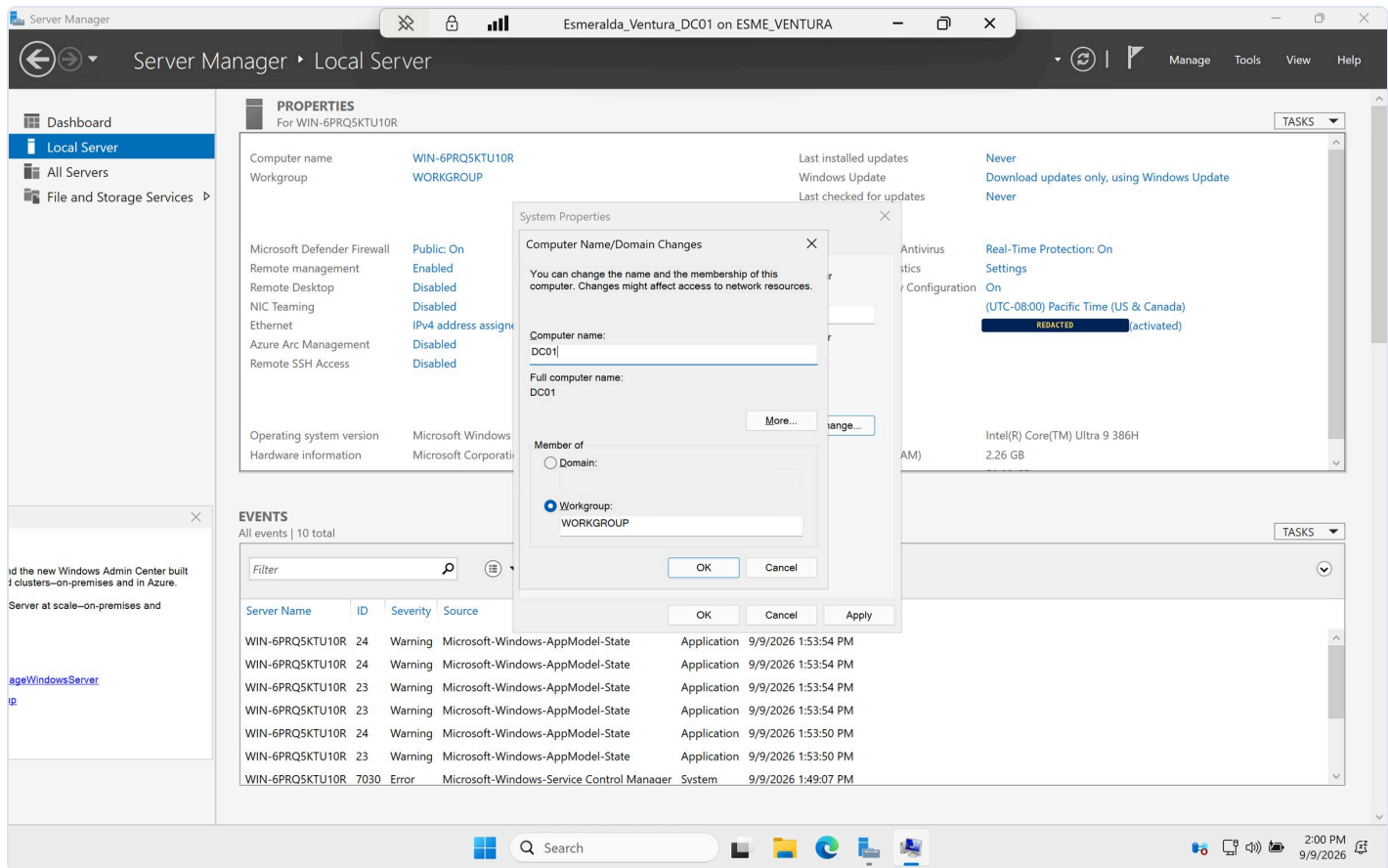


Figure 8. Server Manager, Computer Name/Domain Changes: the default WIN-... name is replaced with DC01 while the server is still in a workgroup, before promotion. The product ID is redacted.

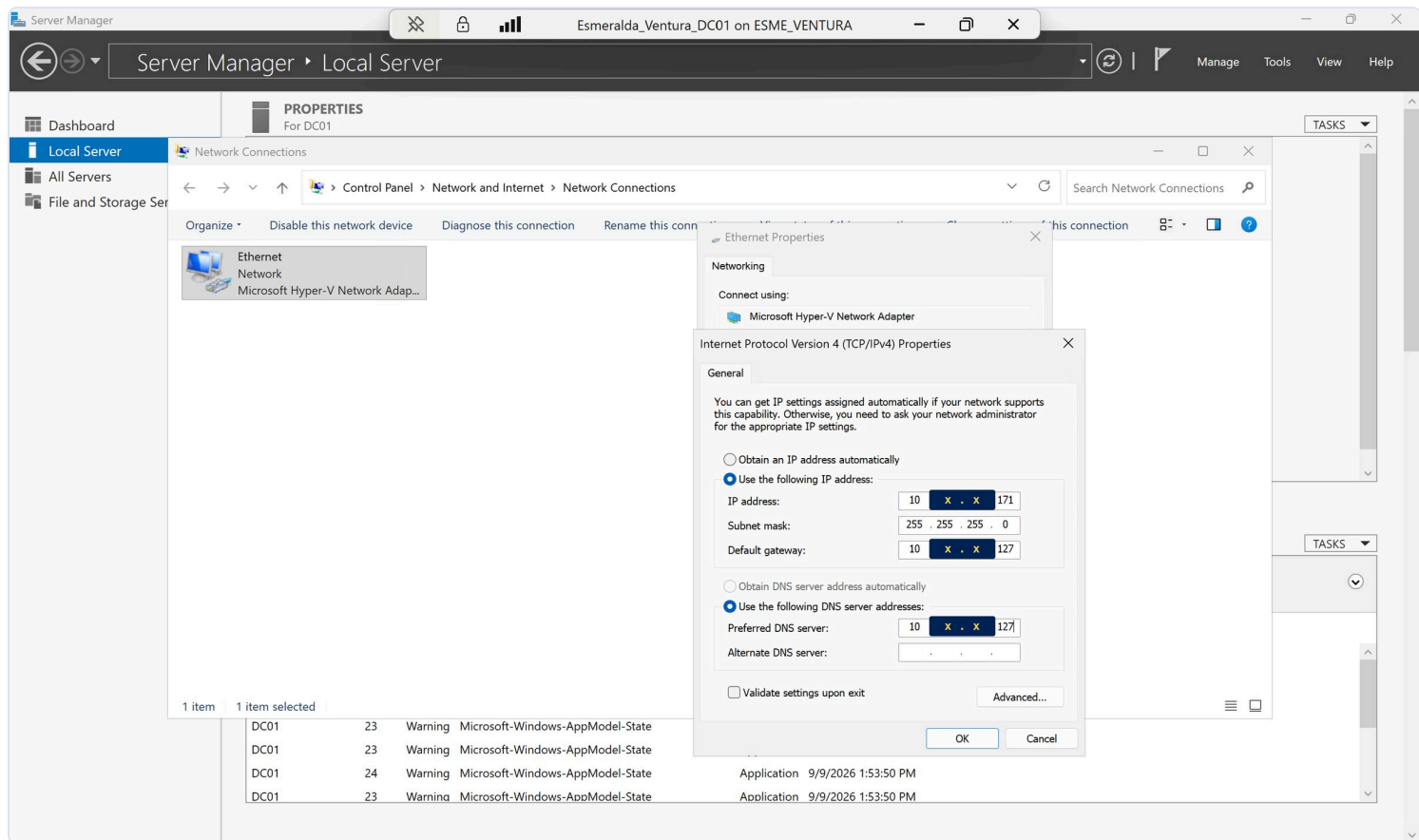


Figure 9. IPv4 properties with Use the following IP address selected: static address, 255.255.255.0 mask, default gateway and preferred DNS server entered manually. Middle octets are masked.

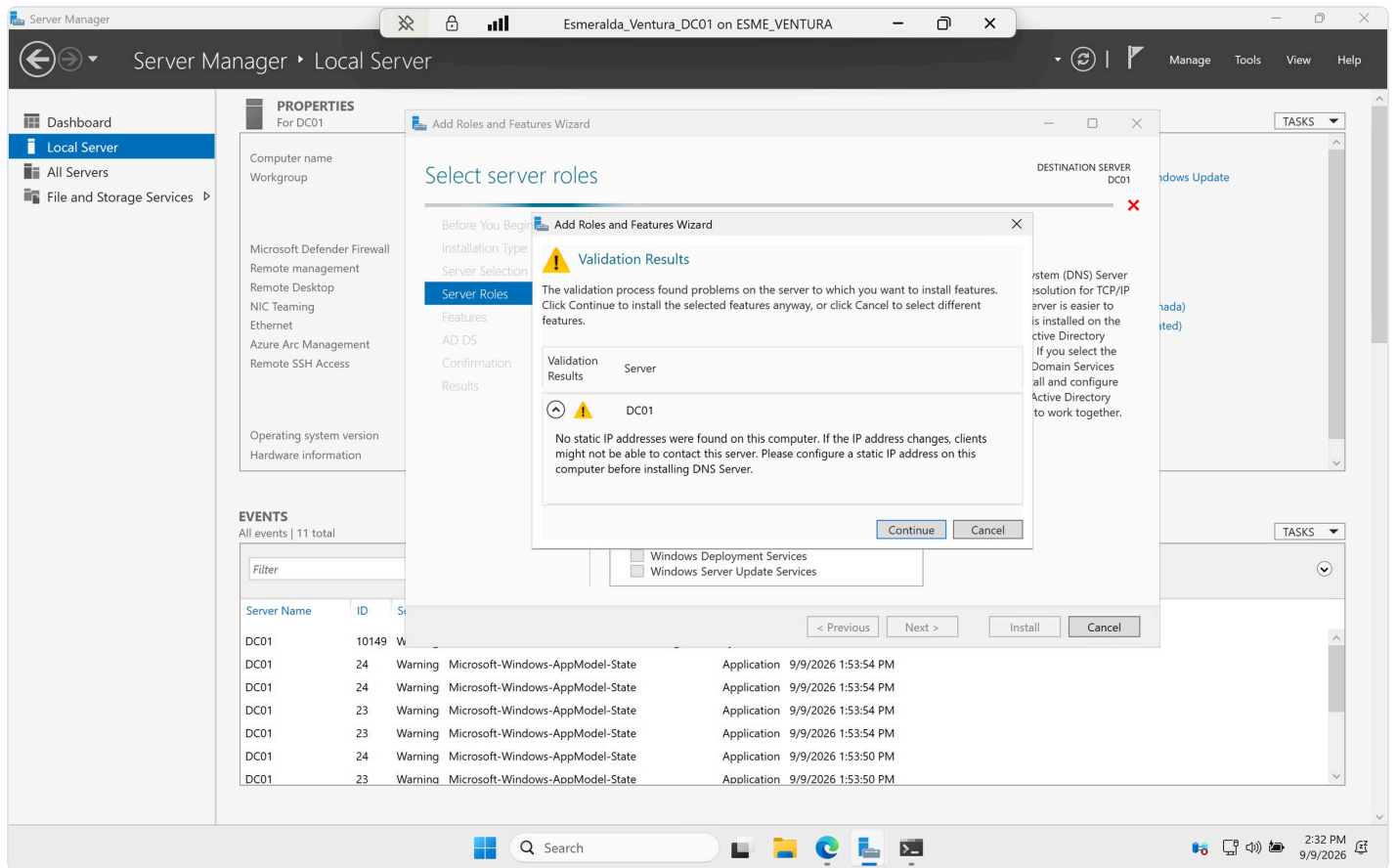


Figure 10. Add Roles and Features raises "No static IP addresses were found on this computer" for DC01, despite the static IPv4 configuration shown in Figure 9.

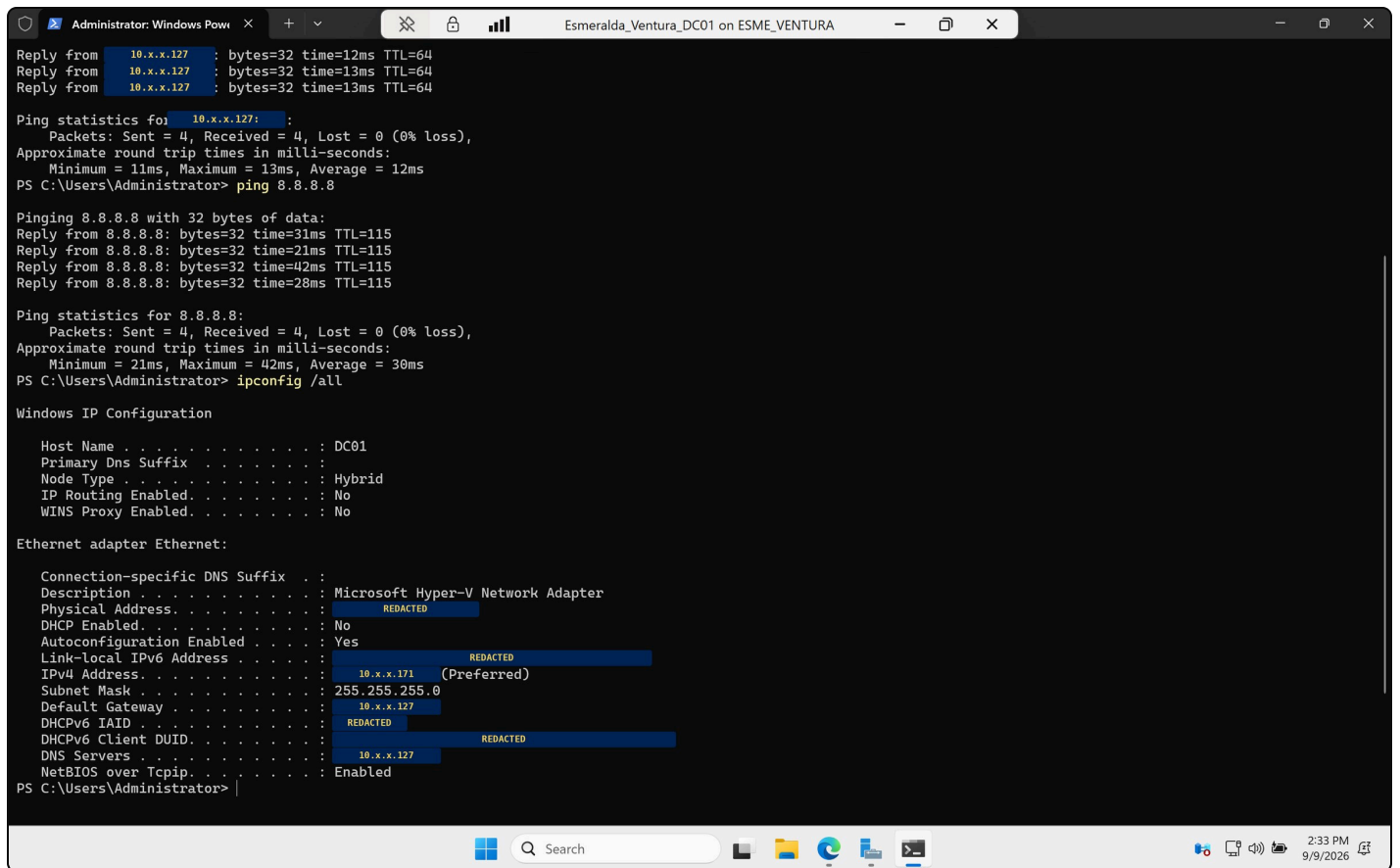


Figure 11. ipconfig /all on DC01: gateway pings succeed, DHCP Enabled: No, and the IPv4 address is marked (Preferred). This evidence justified continuing past the warning. MAC, DHCPv6 and IPv6 values are redacted.

03 Build log: steps 4–5

Step 4: Installing AD DS and creating the forest

The AD DS and DNS roles were installed (Figure 12), then DC01 was promoted as the first domain controller of a new forest (Figures 13–16). Choosing the Windows Server 2025 functional level means the directory can use the newest security features, since there are no older domain controllers to stay compatible with.

Forest parameter	Value
Root domain / NetBIOS name	corp.ventura.local / VENTURA
Functional levels	Windows2025Forest and Windows2025Domain
DC capabilities	DNS server, Global Catalog (writable DC, not RODC)
Database, logs, SYSVOL	C:\WINDOWS\NTDS and C:\WINDOWS\SYSVOL

An expected notice. The prerequisites check warned that a DNS delegation could not be created. That is normal for an internal forest with no parent zone to delegate from, so no action was required.

Step 5: Promotion and post-promotion checks

After the automatic restart, the sign-in screen offered the VENTURA domain (Figure 17). I pointed DC01's preferred DNS server to itself, then verified the domain from PowerShell rather than trusting the wizard (Figures 18–19).

Check	Method	Outcome	Result
DC uses its own DNS	IPv4 settings, <code>ipconfig /all</code>	Preferred DNS is the DC itself (10.x.x.171 and ::1)	✓ Pass
Domain sign-in available	Lock screen after reboot	"Sign in to: VENTURA" offered	✓ Pass
Domain configuration	<code>Get-ADDomain</code>	corp.ventura.local , NetBIOS VENTURA, Windows2025Domain	✓ Pass
Forest and FSMO roles	<code>Get-ADForest</code>	Windows2025Forest; DC01 is schema master, domain naming master and GC	✓ Pass
Name resolution	<code>nslookup</code>	corp.ventura.local resolved by the local DNS server	✓ Pass

Uncropped screenshots for these steps follow as Figures 12–19.

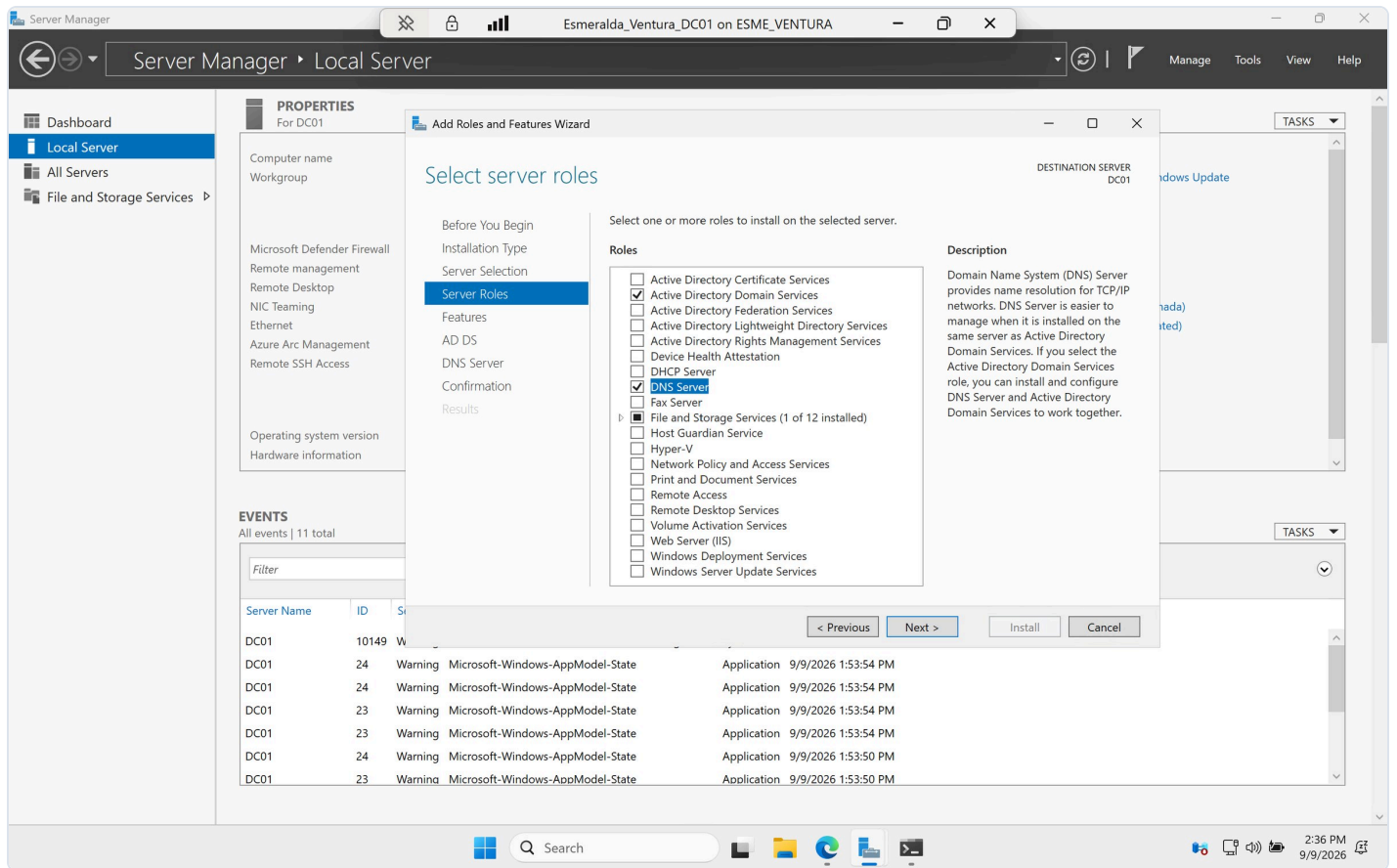


Figure 12. Add Roles and Features on DC01: **Active Directory Domain Services** and **DNS Server** selected, with their management tools.

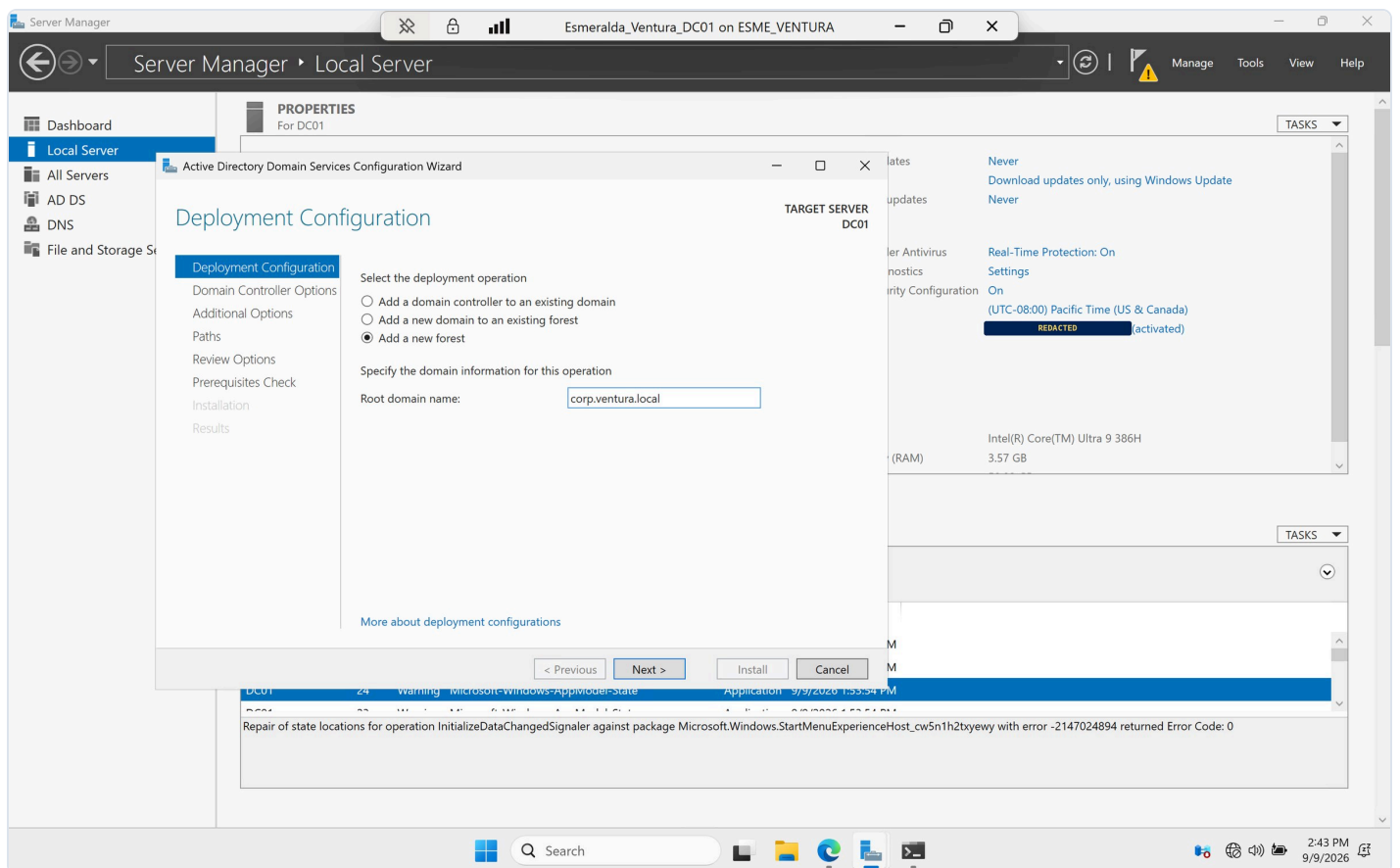


Figure 13. Active Directory Domain Services Configuration Wizard: **Add a new forest** with root domain name **corp.ventura.local**. The product ID in the background is redacted.

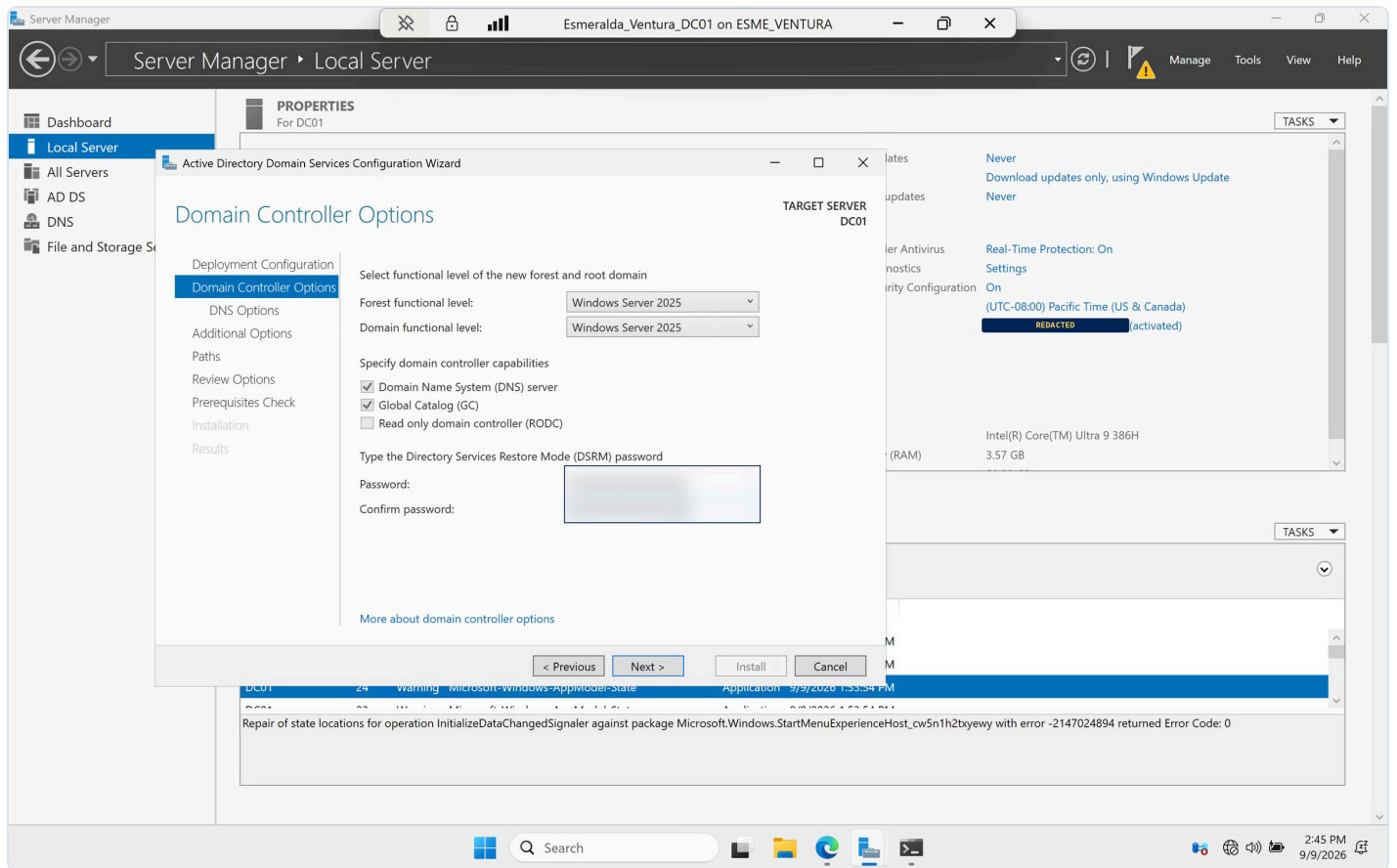


Figure 14. Forest and domain functional level **Windows Server 2025**, with DNS server and Global Catalog selected. The DSRM password fields are blurred.

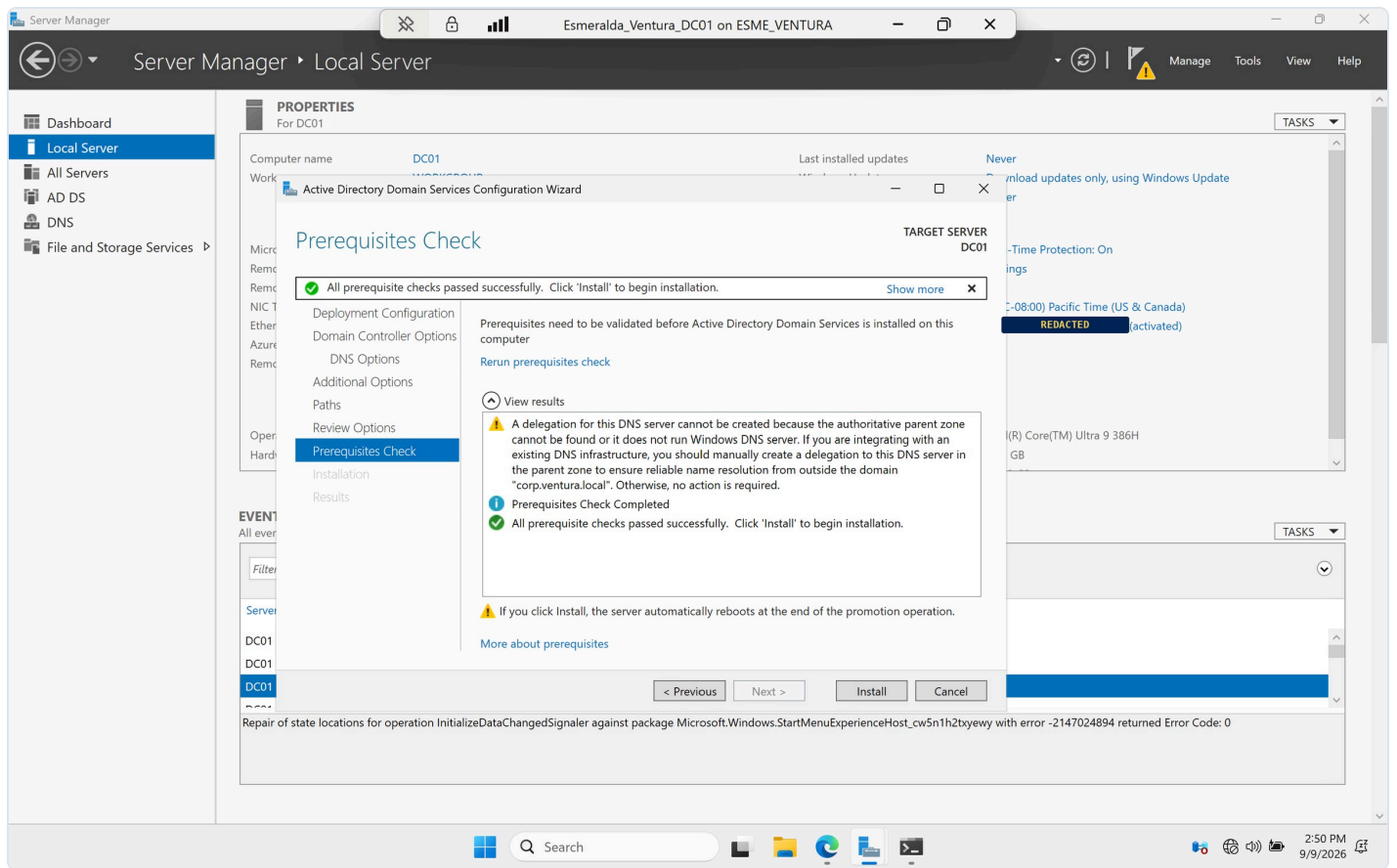


Figure 15. All prerequisite checks passed successfully. The DNS delegation notice is expected for an internal forest with no parent zone.

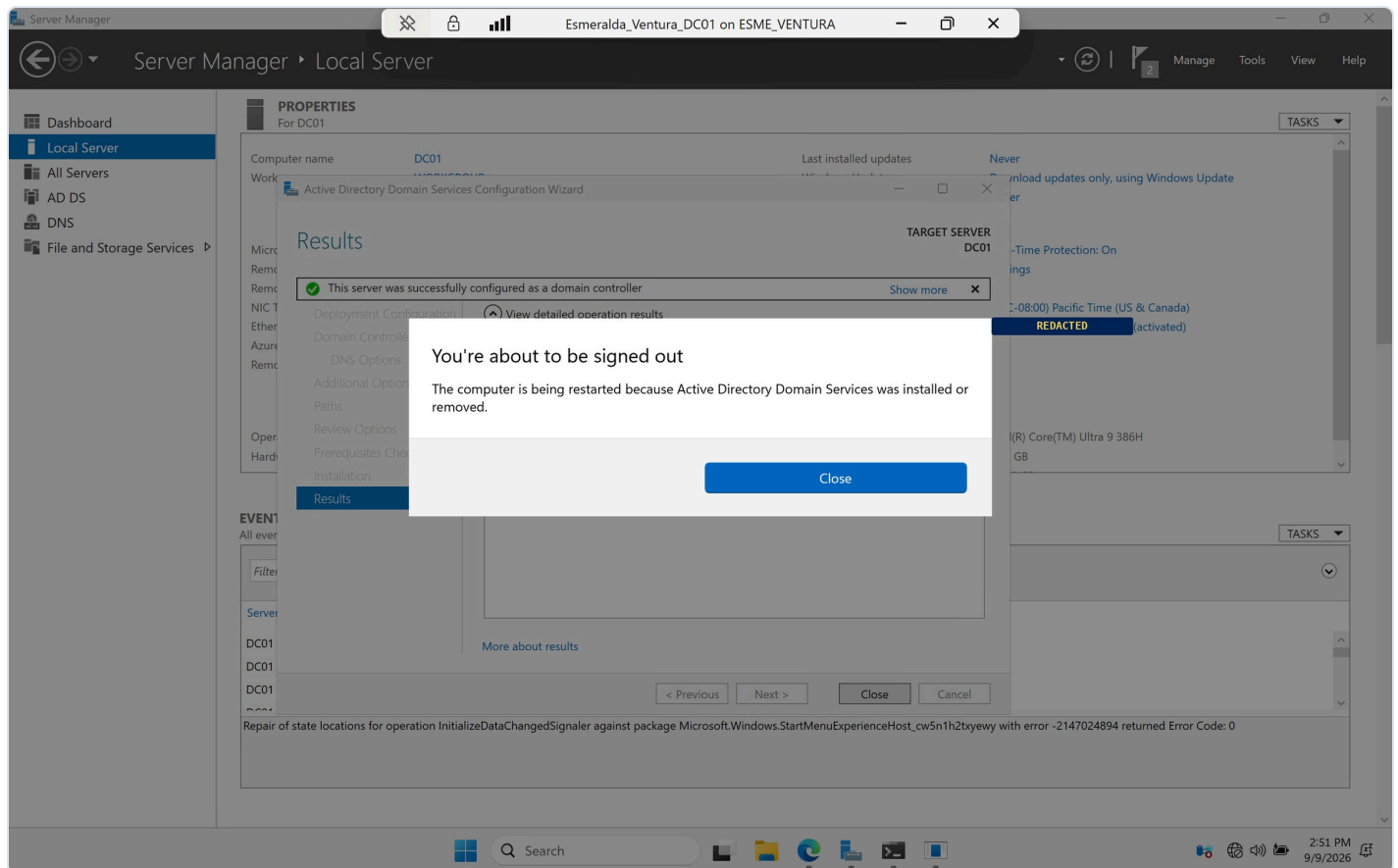


Figure 16. "This server was successfully configured as a domain controller", followed by the automatic sign-out and restart.

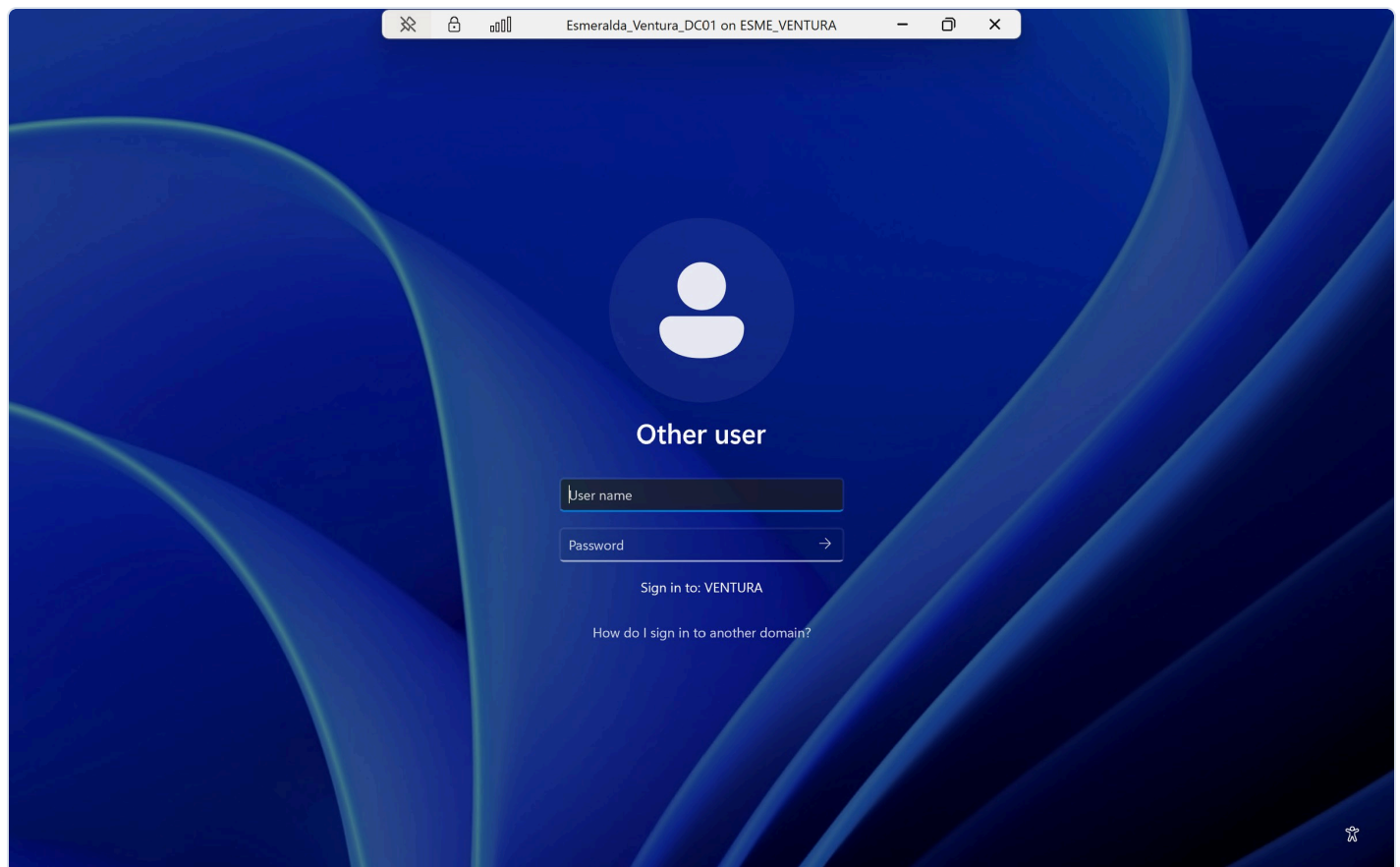


Figure 17. After reboot, the Esmeralda_Ventura_DC01 console shows Sign in to: VENTURA, confirming the new domain is live.

```
Administrator: Windows Powe x + v Esmeralda_Ventura_DC01 on ESME_VENTURA - X
DNS Servers : 10.x.x.171
NetBIOS over Tcpip. : Enabled
PS C:\Users\Administrator> Get-ADDomain

AllowedDNSSuffixes : {}
ChildDomains : {}
ComputersContainer : CN=Computers,DC=corp,DC=ventura,DC=local
DeletedObjectsContainer : CN=Deleted Objects,DC=corp,DC=ventura,DC=local
DistinguishedName : DC=corp,DC=ventura,DC=local
DNSRoot : corp.ventura.local
DomainControllersContainer : OU=Domain Controllers,DC=corp,DC=ventura,DC=local
DomainMode : Windows2025Domain
DomainSID : REDACTED
ForeignSecurityPrincipalsContainer : CN=ForeignSecurityPrincipals,DC=corp,DC=ventura,DC=local
Forest : corp.ventura.local
InfrastructureMaster : DC01.corp.ventura.local
LastLogonReplicationInterval :
LinkedGroupPolicyObjects : {CN={31B2F340-016D-11D2-945F-00C04FB984F9}, CN=Policies, CN=System, DC=corp, DC=ventura, DC=local}
LostAndFoundContainer : CN=LostAndFound, DC=corp, DC=ventura, DC=local
ManagedBy :
Name : corp
NetBIOSName : VENTURA
ObjectClass : domainDNS
ObjectGUID : REDACTED
ParentDomain :
PDCEmulator : DC01.corp.ventura.local
PublicKeyRequiredPasswordRolling : True
QuotasContainer : CN=NTDS Quotas, DC=corp, DC=ventura, DC=local
ReadOnlyReplicaDirectoryServers : {}
ReplicaDirectoryServers : {DC01.corp.ventura.local}
RIDMaster : DC01.corp.ventura.local
SubordinateReferences : {DC=ForestDnsZones, DC=corp, DC=ventura, DC=local, DC=DomainDnsZones, DC=corp, DC=ventura, DC=local, CN=Configuration, DC=corp, DC=ventura, DC=local}
SystemsContainer : CN=System, DC=corp, DC=ventura, DC=local
UsersContainer : CN=Users, DC=corp, DC=ventura, DC=local

PS C:\Users\Administrator>
```

Figure 18. Get-ADDomain : DNSRoot corp.ventura.local, NetBIOSName VENTURA, DomainMode Windows2025Domain, with PDCEmulator and RIDMaster on DC01.corp.ventura.local. The domain SID and ObjectGUID are redacted.

```
Administrator: Windows Powe x + v Esmeralda_Ventura_DC01 on ESME_VENTURA - X
PDCEmulator : DC01.corp.ventura.local
PublicKeyRequiredPasswordRolling : True
QuotasContainer : CN=NTDS Quotas, DC=corp, DC=ventura, DC=local
ReadOnlyReplicaDirectoryServers : {}
ReplicaDirectoryServers : {DC01.corp.ventura.local}
RIDMaster : DC01.corp.ventura.local
SubordinateReferences : {DC=ForestDnsZones, DC=corp, DC=ventura, DC=local, DC=DomainDnsZones, DC=corp, DC=ventura, DC=local, CN=Configuration, DC=corp, DC=ventura, DC=local}
SystemsContainer : CN=System, DC=corp, DC=ventura, DC=local
UsersContainer : CN=Users, DC=corp, DC=ventura, DC=local

PS C:\Users\Administrator> Get-ADForest

ApplicationPartitions : {DC=DomainDnsZones, DC=corp, DC=ventura, DC=local, DC=ForestDnsZones, DC=corp, DC=ventura, DC=local}
CrossForestReferences : {}
DomainNamingMaster : DC01.corp.ventura.local
Domains : {corp.ventura.local}
ForestMode : Windows2025Forest
GlobalCatalogs : {DC01.corp.ventura.local}
Name : corp.ventura.local
PartitionsContainer : CN=Partitions, CN=Configuration, DC=corp, DC=ventura, DC=local
RootDomain : corp.ventura.local
SchemaMaster : DC01.corp.ventura.local
Sites : {Default-First-Site-Name}
SPNSuffixes : {}
UPNSuffixes : {}

PS C:\Users\Administrator> nslookup corp.ventura.local
Server: localhost
Address: ::1

Name: corp.ventura.local
Addresses: REDACTED
10.x.x.171

PS C:\Users\Administrator>
```

Figure 19. Get-ADForest : ForestMode Windows2025Forest, with SchemaMaster, DomainNamingMaster and GlobalCatalogs all on DC01. nslookup corp.ventura.local is answered by the local DNS server.

04 Directory and access model

Automation is only as good as the structure it acts on. I designed the directory so that where an account lives and which groups it belongs to both follow directly from HR data.

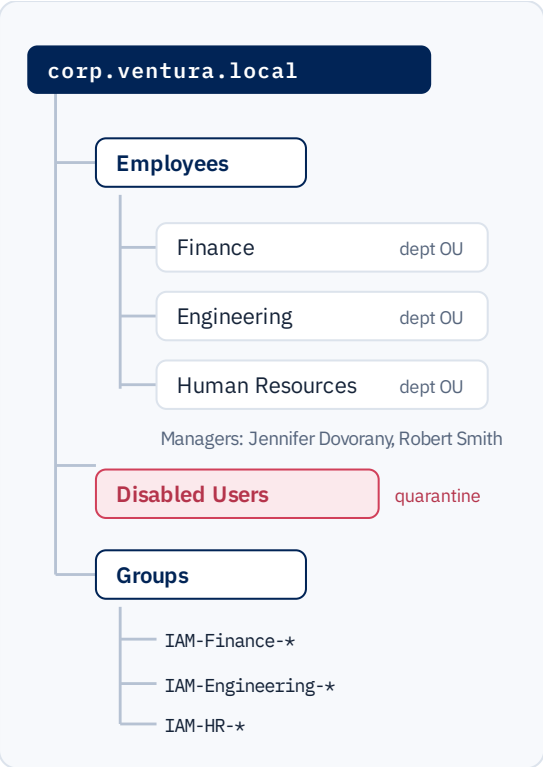


Figure 20. OU design. Department OUs mirror HR departments, and all OUs are protected from accidental deletion.

How access is composed

Desired access is calculated as **department baseline + role overlay**. The department decides the OU and the core groups every member needs. The job title adds role-specific entitlements on top. Both come from one JSON file, so changing policy is a configuration change, not a code change.

Department baseline	Finance	Engineering	HR
Active Directory (validated)			
Groups	IAM-Finance-Users IAM-Finance-SharedDrive	IAM-Engineering-Users IAM-Engineering-SharedDrive	IAM-HR-Users IAM-HR-SharedDrive
Cloud (implemented)			
Entra ID	IAM-Finance-M365	IAM-Engineering-M365	IAM-HR-M365
Intune	IAM-Intune-Corporate-Standard (all departments)		
AVD	Finance-Desktop	Engineering-Desktop	HR-Desktop
Role overlay			
Financial Analyst	IAM-Finance-PowerBI	none	
Cloud Engineer	IAM-Azure-Developers	IAM-Intune-Engineering	
HR Specialist	IAM-HR-Applications	none	

Why a naming prefix is a security control

All automation-owned groups start with `IAM-`. The engine only reads and changes memberships in groups with that prefix. Break-glass, built-in and project groups managed by people stay outside its reach. This keeps automation predictable and makes ownership obvious to anyone browsing the directory.

Why managers are real directory objects. The HR record carries a manager UPN, which the engine resolves to the manager's distinguished name and writes to the `manager` attribute. That attribute powers org charts, approval routing and access reviews. The OUs, groups and manager accounts built for this lab are shown full size in Figures 21–24.

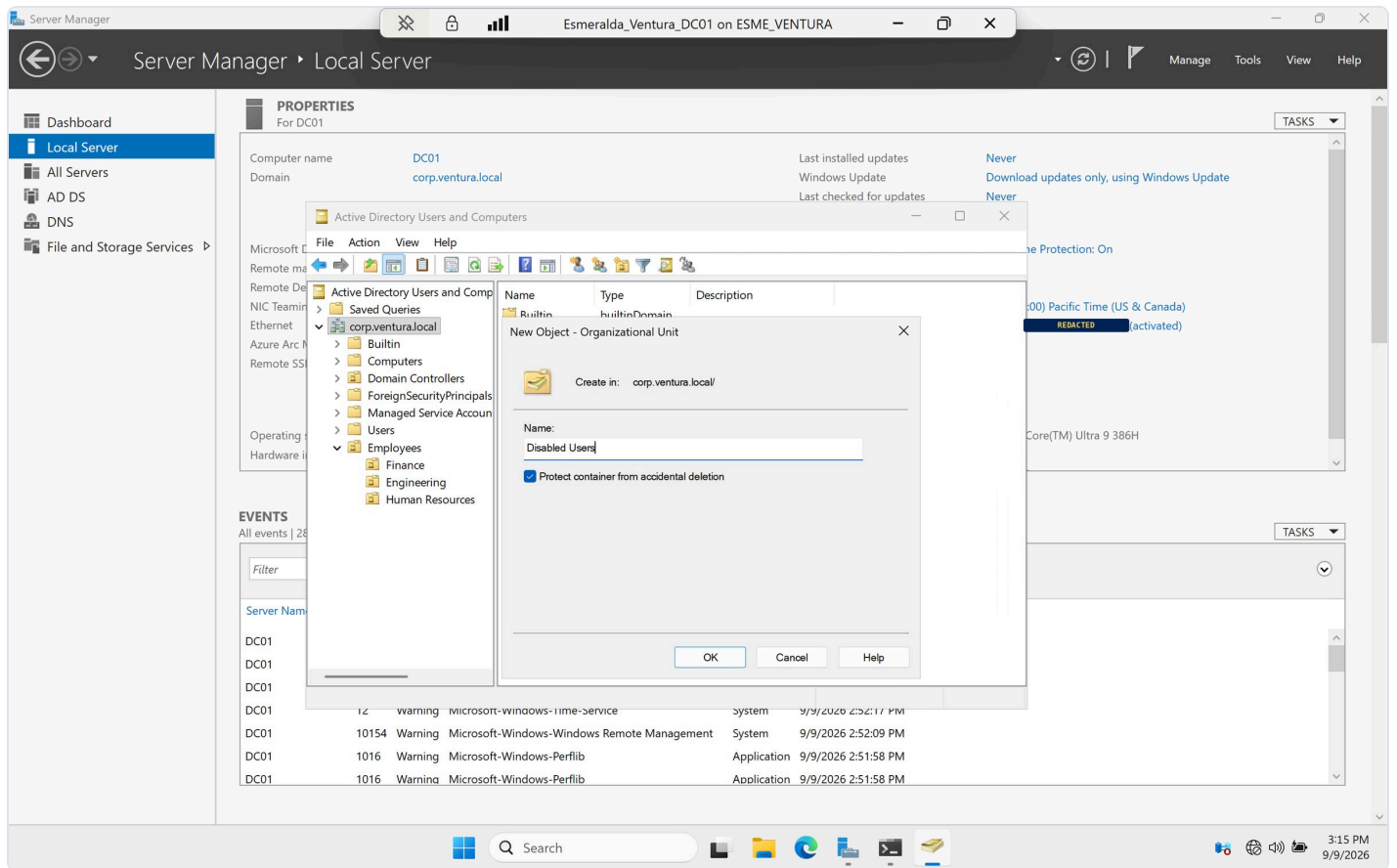


Figure 21. Active Directory Users and Computers: creating **Disabled Users** at the domain root with **Protect container from accidental deletion** enabled. Finance, Engineering and Human Resources already exist under Employees.

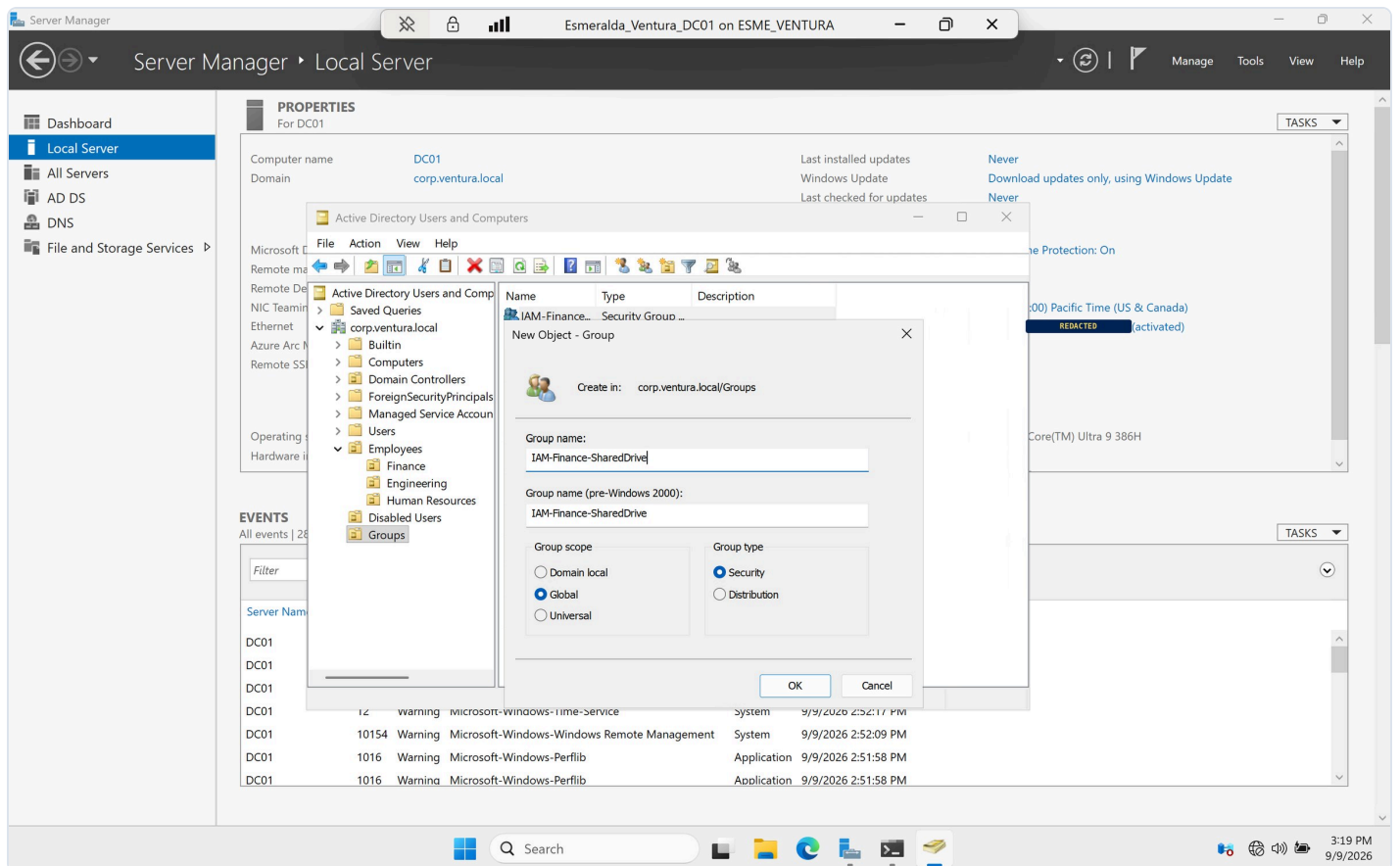


Figure 22. Creating **IAM-Finance-SharedDrive** in the Groups OU with group scope **Global** and group type **Security**.

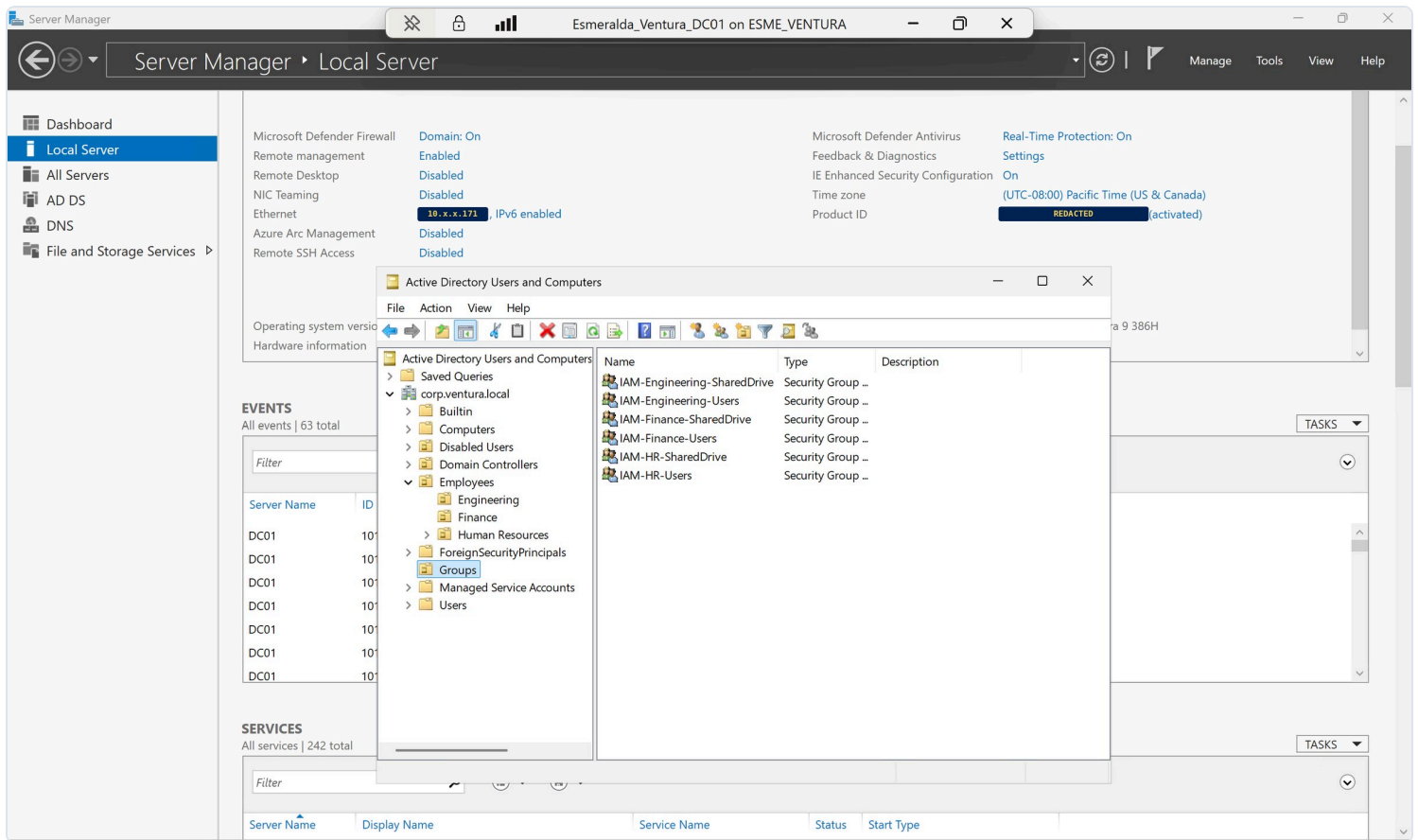


Figure 23. The Groups OU holding all six automation-managed groups: **IAM-Engineering**, **IAM-Finance** and **IAM-HR**, each with Users and SharedDrive variants.

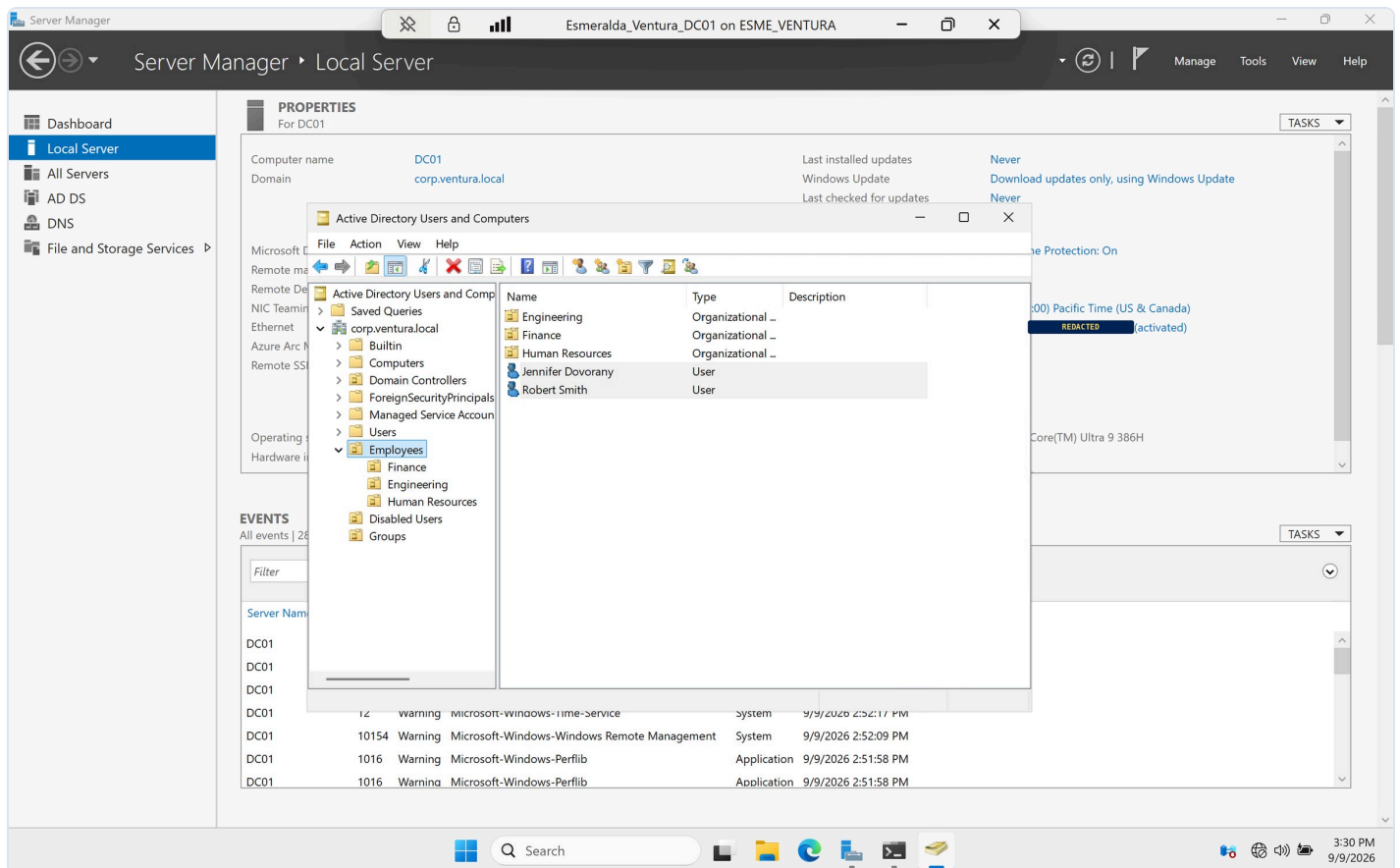


Figure 24. The Employees OU containing the three department OUs and the manager accounts **Jennifer Dovorany** and **Robert Smith**.

05 Inside the automation engine

The engine is a single 983-line script organized into 30 functions. Each layer has one job, so connectors can be added or swapped without touching the lifecycle logic.

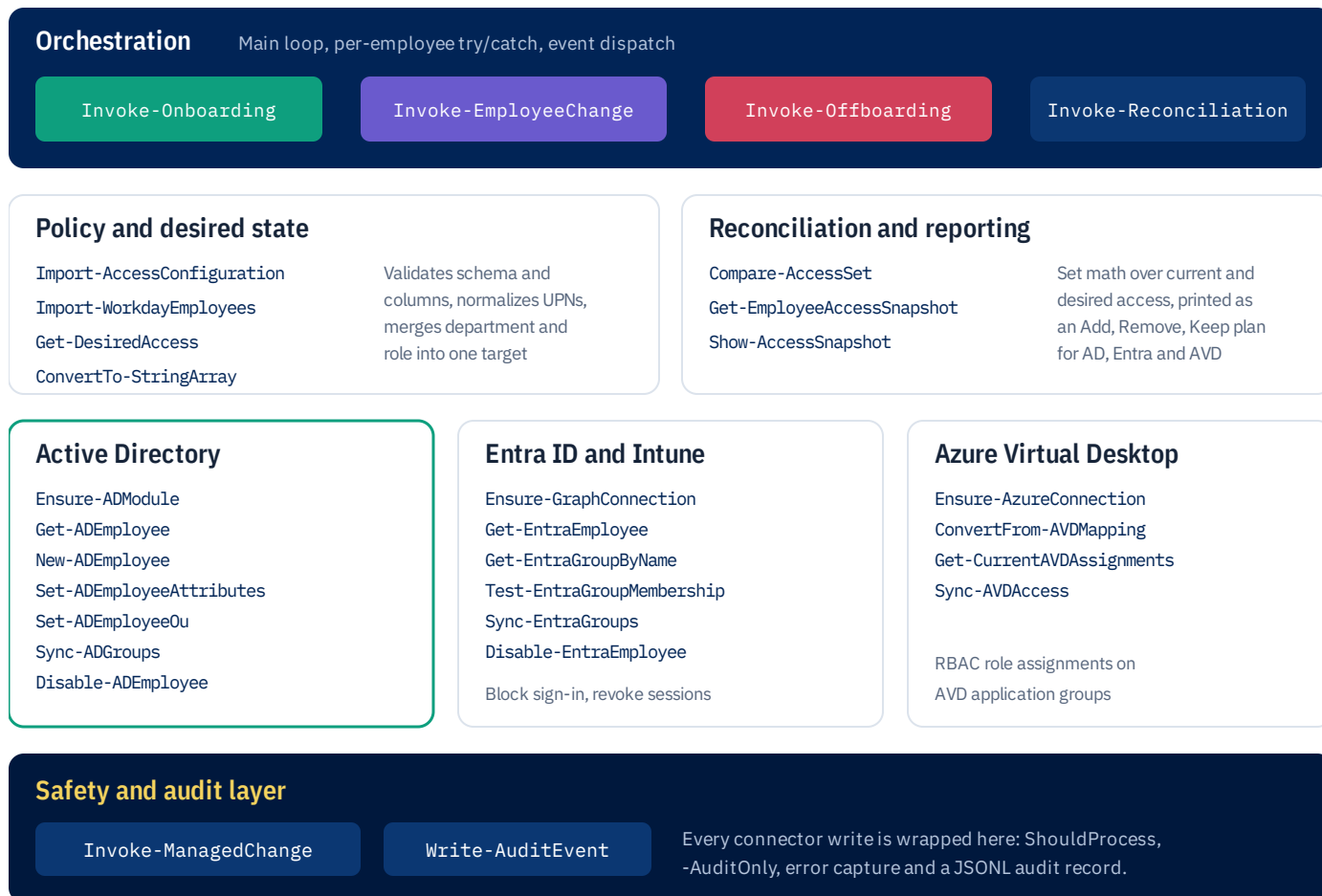


Figure 25. Function map of `Invoke-EmployeeIdentityLifecycle.ps1`, showing all 30 functions grouped by layer.

Lifecycle event	Active Directory actions, in order
Hire	Create the user in the department OU with a must-change-at-logon password, set title and department, link the manager, place in the OU, add baseline and role groups.
RoleChange Transfer	Update attributes and manager, move the OU if the department changed, then add missing groups and remove groups no longer justified.
Termination	Disable the account, remove every managed IAM- membership, move to the Disabled Users OU. In the cloud, also block sign-in and revoke sessions.

05 The reconciliation core

The heart of the engine is a small, testable function that treats access as sets. The diagrams use real data from the lab tests: the role change kept access, and the transfer swapped it.

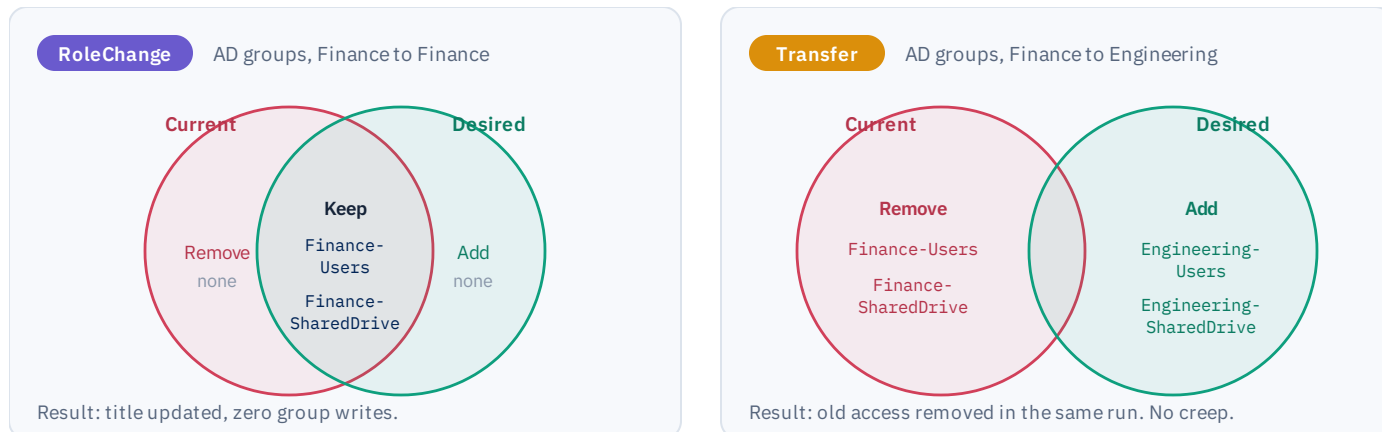


Figure 26. Set reconciliation using real test data. Groups without the **IAM-** prefix, such as Domain Users, are filtered out before comparison and never touched.

```
function Compare-AccessSet {  
    [CmdletBinding()]  
    param(  
        [string[]]$Current,  
        [string[]]$Desired  
    )  
    $currentNormalized = @($Current | Sort-Object -Unique)  
    $desiredNormalized = @($Desired | Sort-Object -Unique)  
  
    [pscustomobject]@{  
        Add = @($desiredNormalized |  
            Where-Object { $_ -notin $currentNormalized })  
        Remove = @($currentNormalized |  
            Where-Object { $_ -notin $desiredNormalized })  
        Keep = @($desiredNormalized |  
            Where-Object { $_ -in $currentNormalized })  
    }  
}
```

Pure function with no side effects. It is reused for AD, Entra, Intune and AVD, and is easy to unit test.

```
function Invoke-ManagedChange {  
    param($Employee, $System, $Action, $ScriptBlock)  
  
    if ($AuditOnly) { # plan only, never write  
        Write-AuditEvent ... -Result Planned  
        return  
    }  
    $what = "$System - $Action"  
    if ($PSCmdlet.ShouldProcess($Target, $what)) {  
        try {  
            & $ScriptBlock  
            Write-AuditEvent ... -Result Success  
        }  
        catch {  
            Write-AuditEvent ... -Result Failed ` -  
                -Details $_.Exception.Message  
            throw  
        }  
    }  
    else { Write-AuditEvent ... -Result Skipped }  
}
```

Condensed for print. Every connector passes its change in as a script block, so all writes share one safety path.

05 Data contracts and desired-state logic

HR input contract

The engine refuses input that doesn't meet the contract. A missing required column, an empty file or an unknown lifecycle event stops the run before any change is attempted.

Column	Required	Used for
EmployeeId	Yes	Immutable match key (employeeID)
FirstName , LastName	Yes	Name, display name
UserPrincipalName	Yes	Sign-in name, lower-cased
SamAccountName	No	Derived from UPN if blank
Department , JobTitle	Yes	OU and group calculation
ManagerUpn	Yes	Resolved to manager DN
Status	Yes	Terminated or Inactive forces offboarding
LifecycleEvent	Yes	Hire, RoleChange, DepartmentTransfer, Termination, Reconcile
Location , EmploymentType	No	Carried for future policy rules

Test records used

Event	Dept / title	Manager
Hire	Finance Financial Analyst	Jennifer Dovorany
RoleChange	Finance Cloud Engineer	Jennifer Dovorany
Transfer	Engineering Cloud Engineer	Robert Smith
Termination	Engineering Cloud Engineer	Robert Smith

All four records share employee ID `10001` (Sarah Johnson, `sjohnson`). One identity moves through the full lifecycle, exactly as a real employee would.

Desired state for leavers

```
if ($Employee.LifecycleEvent -eq 'Termination' -or
    $Employee.Status -match 'Terminated|Inactive') {
    return [pscustomobject]@{
        OuPath = $Config.Organization.DisabledUsersOu
        ADGroups = @(); EntraGroups = @()
        IntuneGroups = @(); AVD = @()
    }
}
```

From `Get-DesiredAccess`. For a leaver, the target is empty everywhere. A `Status` of `Terminated` triggers this even if the event column is wrong, which is defense in depth.

Scoped group sync

```
$prefix = $Config.Organization.ManagedADGroupPrefix
$current = @(Get-ADPrincipalGroupMembership $AdUser |
    Where-Object { $_.Name -like "$prefix*" } |
    Select-Object -ExpandProperty Name)

$diff = Compare-AccessSet -Current $current `
    -Desired $DesiredGroups
foreach ($g in $diff.Add) { Add-ADGroupMember ... }
foreach ($g in $diff.Remove) { Remove-ADGroupMember ... }
```

Condensed from `Sync-ADGroups`. The prefix filter is applied before the diff, so unmanaged groups can't appear in the Remove list.

Access policy excerpt

```
"Organization": {
  "InitialPassword": "[REDACTED: lab-only secret, see Security section]",
  "DisabledUsersOu": "OU=Disabled Users,DC=corp,DC=ventura,DC=local",
  "ManagedADGroupPrefix": "IAM-",
  "ManagedAVDApplicationGroups": [ "rg-avd|Finance-Desktop", "rg-avd|Engineering-Desktop", "rg-avd|HR-Desktop" ]
}
```

05 Tooling and deployment to the lab

The engine, policy and test cases were written in VS Code (Figures 27–28), then staged on DC01 under `C:\Lab` (Figure 29). The `Audit` folder did not exist beforehand. The engine created it on its first run (Figure 30), which confirms the logging path works on a clean system.

Project inventory

File	Size	Purpose
<code>Invoke-EmployeeIdentityLifecycle.ps1</code>	34 KB	The engine: 983 lines, 30 functions, all lifecycle logic and connectors
<code>AccessMappings.json</code>	3 KB	Access policy: department baselines, role overlays, OU paths, managed prefixes
<code>EmployeeLifeCycleTestCases\</code>	4 files	Workday-style CSV records for Hire, RoleChange, DepartmentTransfer and Termination
<code>Reset-TestEmployee.ps1</code>	1 KB	Test harness: deletes only the test account and verifies the lab is ready for Test 1
<code>README.md</code>	5 KB	Lab topology, access model, test order and verification commands
<code>Audit\EmployeeIdentityAudit.jsonl</code>	runtime	Append-only audit trail created by the engine

Audit record schema

```
{ "Timestamp": "2026-09-10T16:23:07.000Z", "EmployeeId": "10001",  
  "UserPrincipalName": "sarah.johnson@corp.ventura.local", "LifecycleEvent": "Hire",  
  "System": "ActiveDirectory", "Action": "Add group 'IAM-Finance-Users'",  
  "Result": "Success", "Details": "", "RunBy": "Administrator", "Computer": "DC01" }
```

Representative record built from the fields written by `Write-AuditEvent`. The timestamp shown is illustrative. One compressed JSON object is written per line, a format SIEM tools can ingest directly.

Uncropped screenshots follow as Figures 27–30.

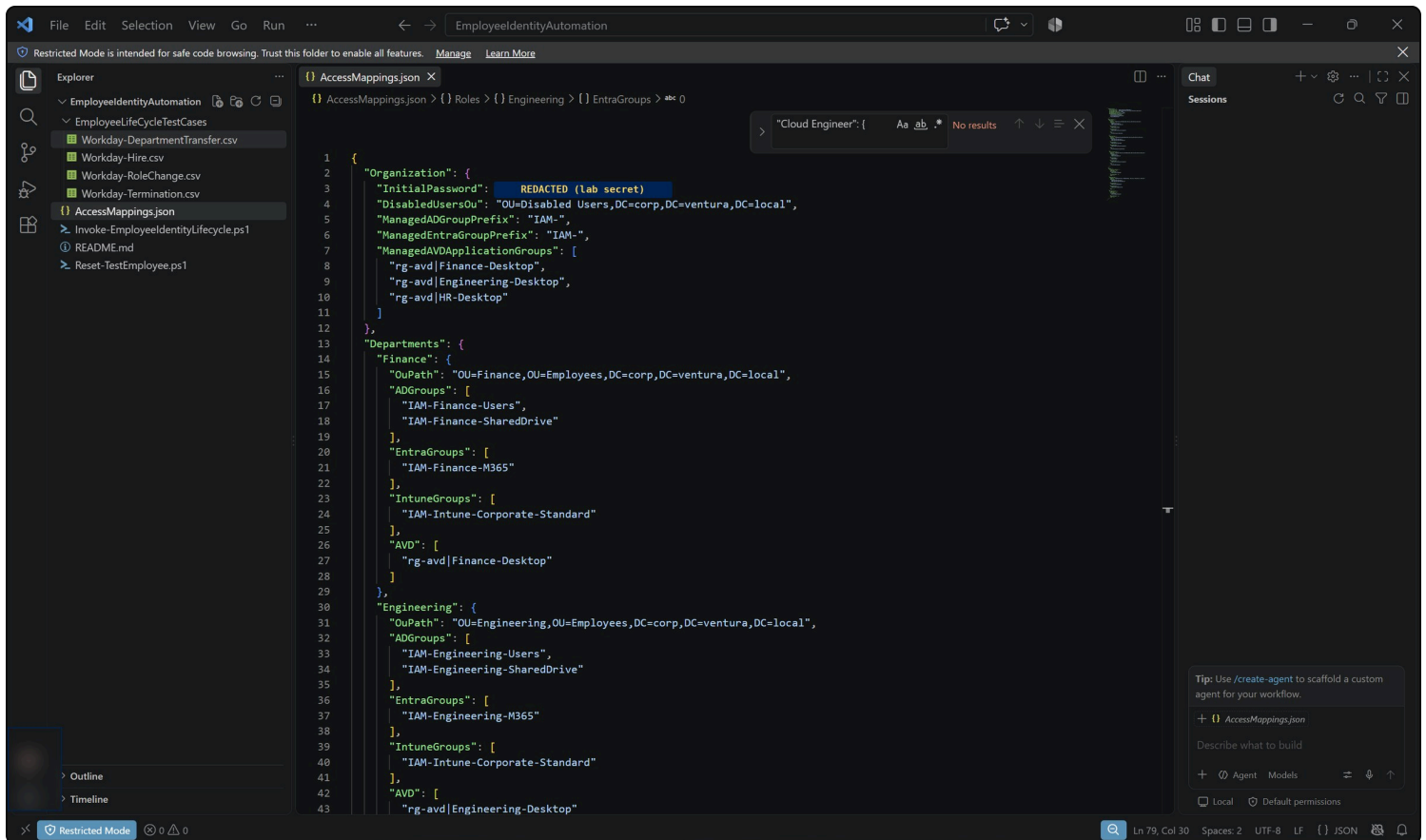


Figure 27. `AccessMappings.json` : organization settings (quarantine OU, **IAM**-prefix, AVD application groups) and the Finance and Engineering department baselines. The initial password is redacted.

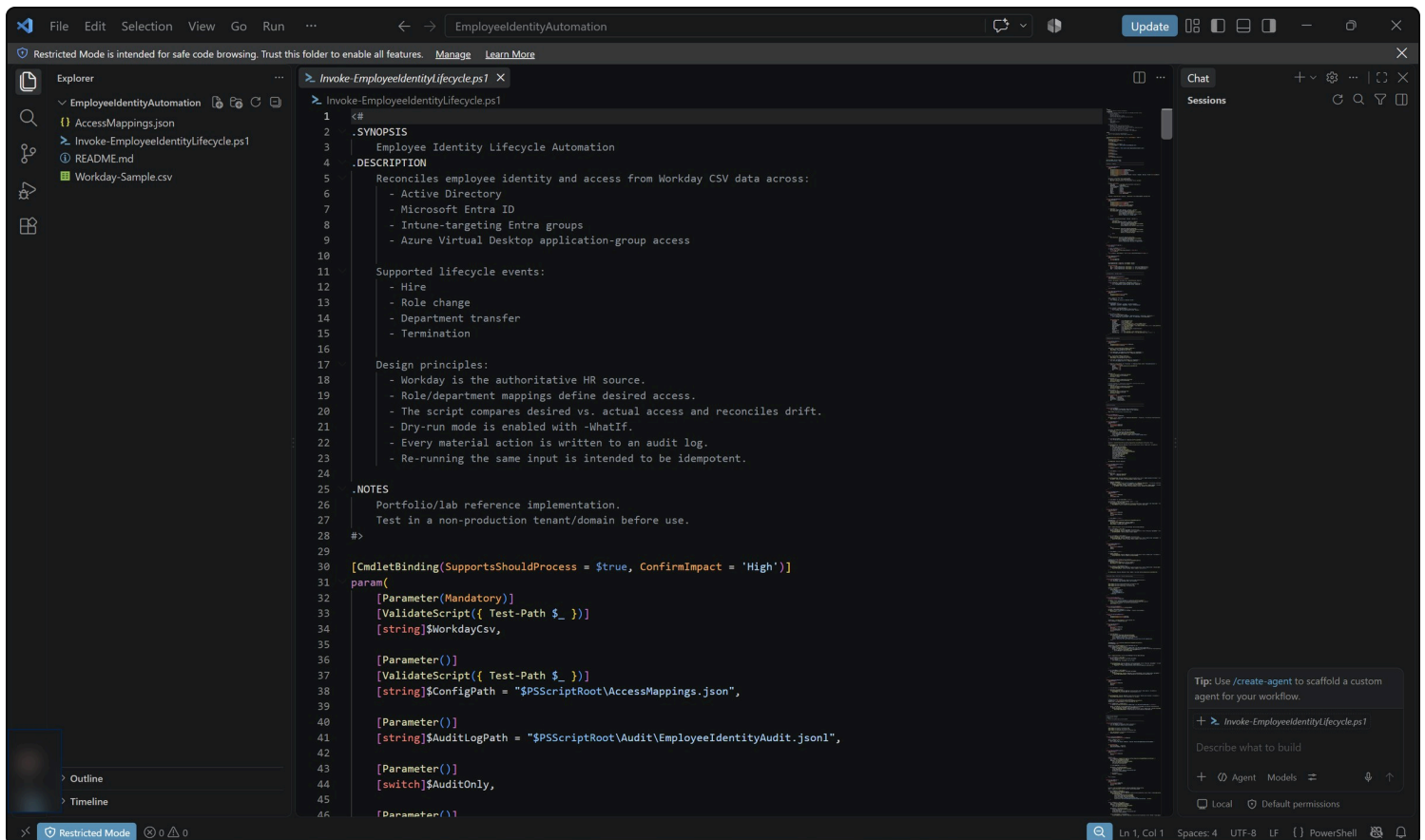


Figure 28. `Invoke-EmployeeIdentityLifecycle.ps1` : comment-based help listing the design principles, and the `CmdletBinding(SupportsShouldProcess = $true, ConfirmImpact = 'High')` parameter block.

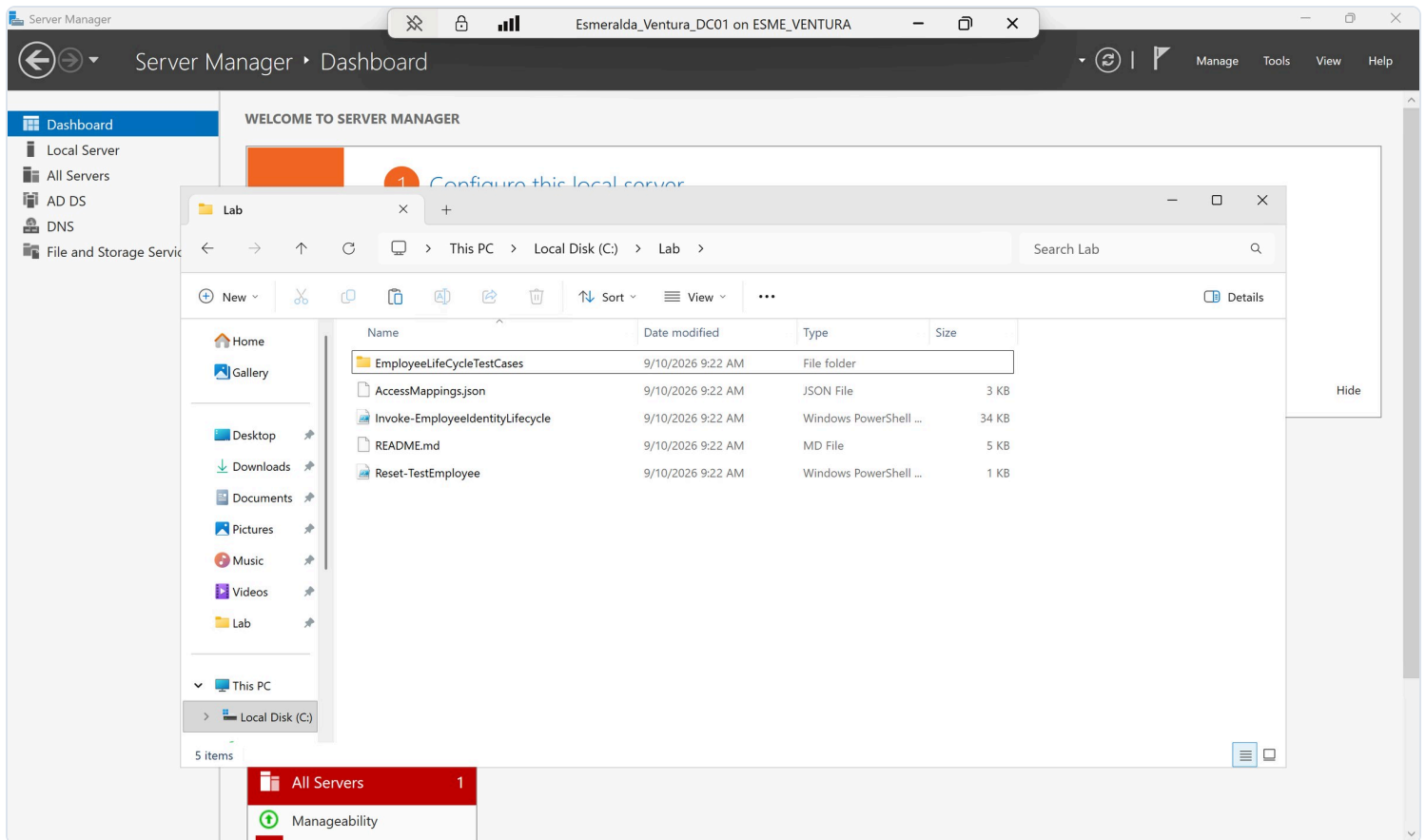


Figure 29. C:\Lab on DC01 before the first run: the test case folder, `AccessMappings.json`, the engine script, `README.md` and `Reset-TestEmployee.ps1`.

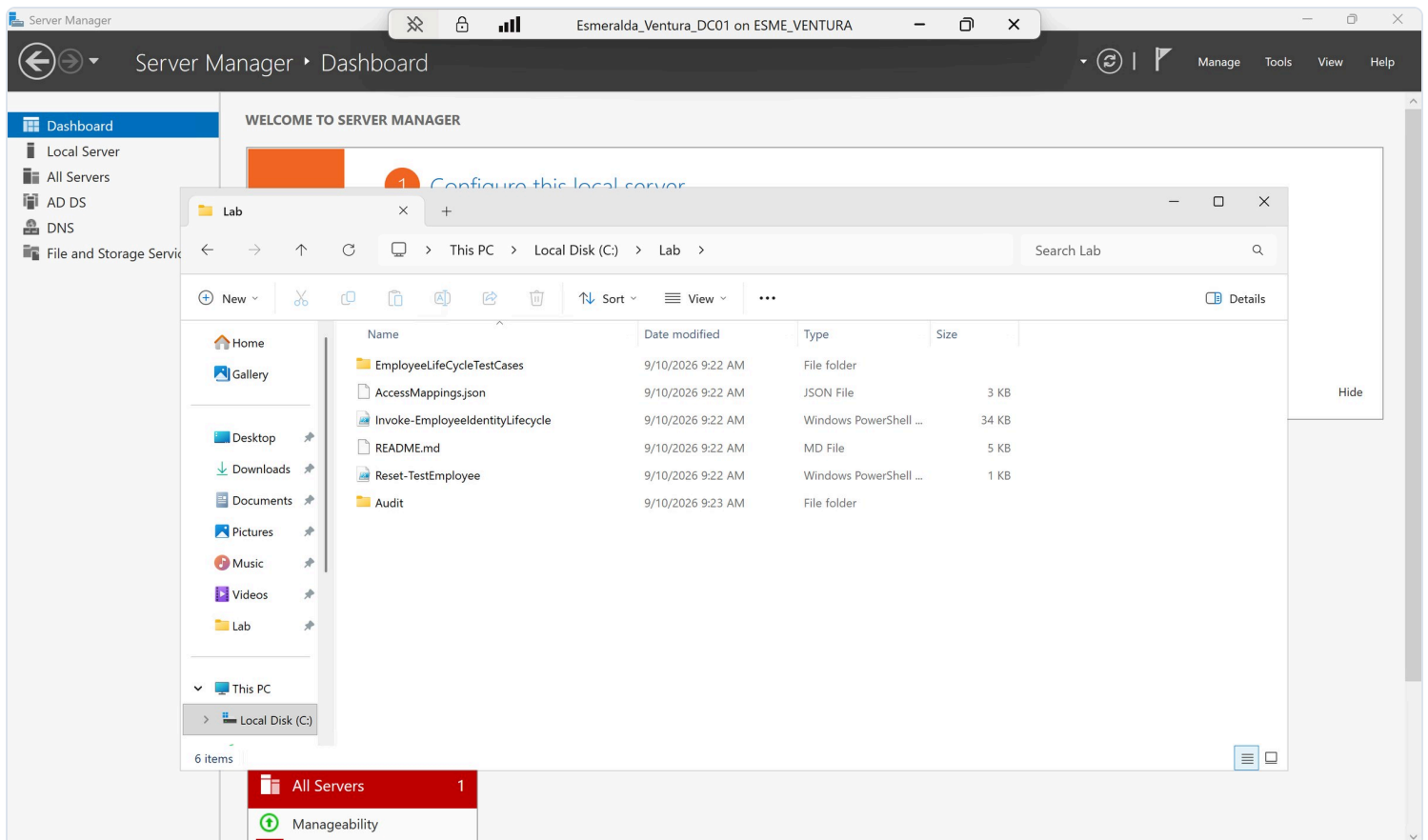


Figure 30. C:\Lab after Test 1: the **Audit** folder now exists, created automatically by the engine.

06 Test campaign: one employee, four lifecycle events

Sarah Johnson was taken from hire to termination with four HR records. After each run, her account was checked twice: from PowerShell with `Get-ADUser`, and visually in Active Directory Users and Computers.

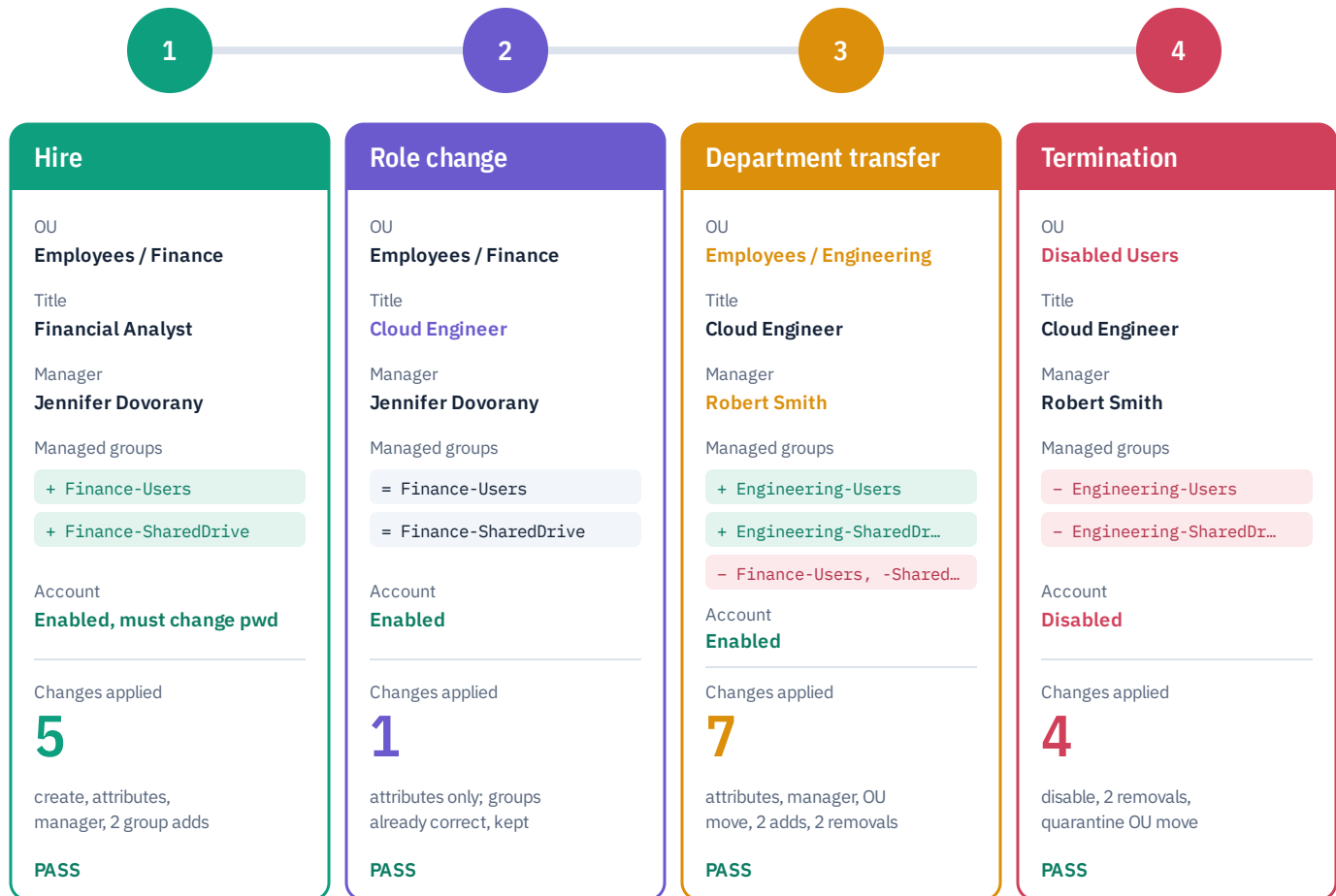


Figure 31. Sarah Johnson's state after each run. Colored values show what changed at that step. Group names drop the `IAM-` prefix for space.

Between full campaign runs, `Reset-TestEmployee.ps1` deletes only the test account and verifies the deletion. Managers, groups and OUs are left untouched, so the four tests can be repeated from a known clean state.

How every run was verified

```
Get-ADUser sjohnson -Properties Department,Title,Manager |  
    Select-Object Name,Department,Title,Manager,DistinguishedName,Enabled  
Get-ADPrincipalGroupMembership sjohnson |  
    Select-Object -ExpandProperty Name | Sort-Object
```

Run immediately after each test. The same state was then checked visually in ADUC, giving two independent views of every result.

Hire: onboarding a new Financial Analyst

HR record: employee 10001 Sarah Johnson, Finance, Financial Analyst, manager Jennifer Dovorany, event **Hire**. Run with `Workday-Hire.csv` and `-SkipCloud`.

Reconciliation plan printed by the engine

System	Add	Remove	Keep
AD	IAM-Finance-SharedDrive , IAM-Finance-Users	none	none
Entra	IAM-Finance-M365 , IAM-Finance-PowerBI , IAM-Intune-Corporate-Standard	none	none
AVD	rg-avd Finance-Desktop	none	none

Changes executed (5)

```
[SUCCESS] ActiveDirectory :: Create user :: sarah.johnson@corp.ventura.local
[SUCCESS] ActiveDirectory :: Update employee attributes :: sarah.johnson@corp.ventura.local
[SUCCESS] ActiveDirectory :: Update manager :: sarah.johnson@corp.ventura.local
[SUCCESS] ActiveDirectory :: Add group 'IAM-Finance-SharedDrive' :: sarah.johnson@corp.ventura.local
[SUCCESS] ActiveDirectory :: Add group 'IAM-Finance-Users' :: sarah.johnson@corp.ventura.local
```

Verification output

Department / Title	Finance / Financial Analyst
Manager	CN=Jennifer Dovorany,OU=Employees,DC=corp,DC=ventura,DC=local
DistinguishedName	CN=Sarah Johnson,OU=Finance,OU=Employees,DC=corp,DC=ventura,DC=local
Enabled	True
Groups	Domain Users, IAM-Finance-SharedDrive, IAM-Finance-Users

Uncropped evidence follows: console output (Figure 32) and Active Directory Users and Computers (Figures 33–34).

```
Administrator: Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PowerShellLatest

PS C:\Lab> .\Invoke-EmployeeIdentityLifecycle.ps1 -WorkdayCsv .\EmployeeLifeCycleTestCases\Workday-Hire.csv -ConfigPath .\AccessMappings.json -SkipCloud

Processing 10001 :: sarah.johnson@corp.ventura.local :: Hire

=====
ACCESS RECONCILIATION: sarah.johnson@corp.ventura.local
Event: Hire
Department: Finance
Role: Financial Analyst
=====

AD
Add: IAM-Finance-SharedDrive, IAM-Finance-Users
Remove:
Keep:

Entra
Add: IAM-Finance-M365, IAM-Finance-PowerBI, IAM-Intune-Corporate-Standard
Remove:
Keep:

AVD
Add: rg-avd|Finance-Desktop
Remove:
Keep:

[SUCCESS] ActiveDirectory :: Create user :: sarah.johnson@corp.ventura.local
[SUCCESS] ActiveDirectory :: Update employee attributes :: sarah.johnson@corp.ventura.local
[SUCCESS] ActiveDirectory :: Update manager :: sarah.johnson@corp.ventura.local
[SUCCESS] ActiveDirectory :: Add group 'IAM-Finance-SharedDrive' :: sarah.johnson@corp.ventura.local
[SUCCESS] ActiveDirectory :: Add group 'IAM-Finance-Users' :: sarah.johnson@corp.ventura.local
PS C:\Lab> Get-ADUser sjohnson -Properties Department,Title,Manager | Select-Object Name,Department,Title,Manager,DistinguishedName,Enabled; Get-ADPrincipalGroupMembership sjohnson | Select-Object -Exp
andProperty Name | Sort-Object

Name           : Sarah Johnson
Department     : Finance
Title          : Financial Analyst
Manager        : CN=Jennifer Dovorany,OU=Employees,DC=corp,DC=ventura,DC=local
DistinguishedName : CN=Sarah Johnson,OU=Finance,OU=Employees,DC=corp,DC=ventura,DC=local
Enabled        : True

Domain Users
IAM-Finance-SharedDrive
IAM-Finance-Users

PS C:\Lab> |
```

Figure 32. Hire run on Esmeralda_Ventura_DC01 : the reconciliation plan for AD, Entra and AVD, five Active Directory changes marked [SUCCESS], and the Get-ADUser verification showing the new account and its groups.

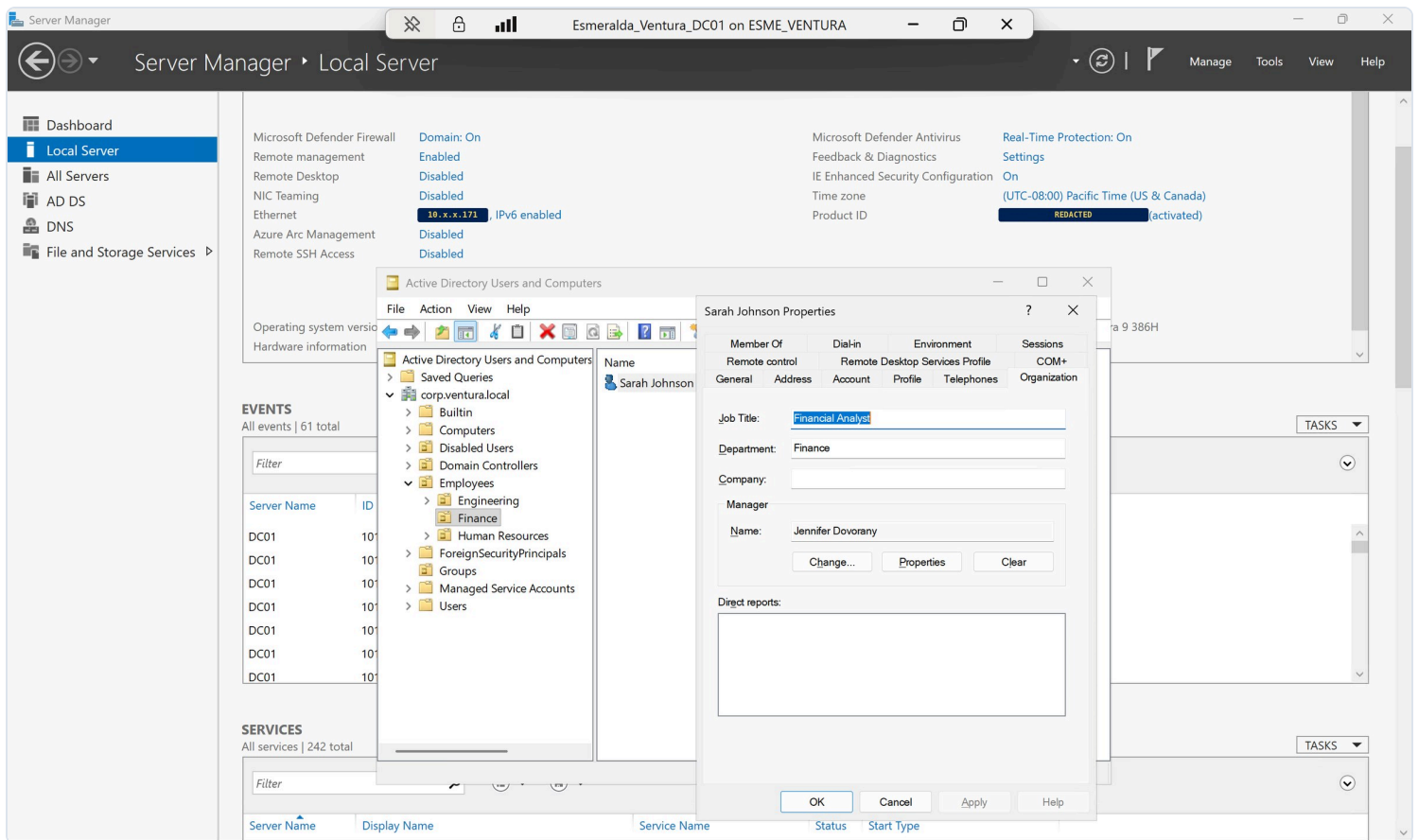


Figure 33. Sarah Johnson properties: Job Title **Financial Analyst**, Department **Finance**, Manager **Jennifer Dovorany**. The account is listed in the Finance OU.

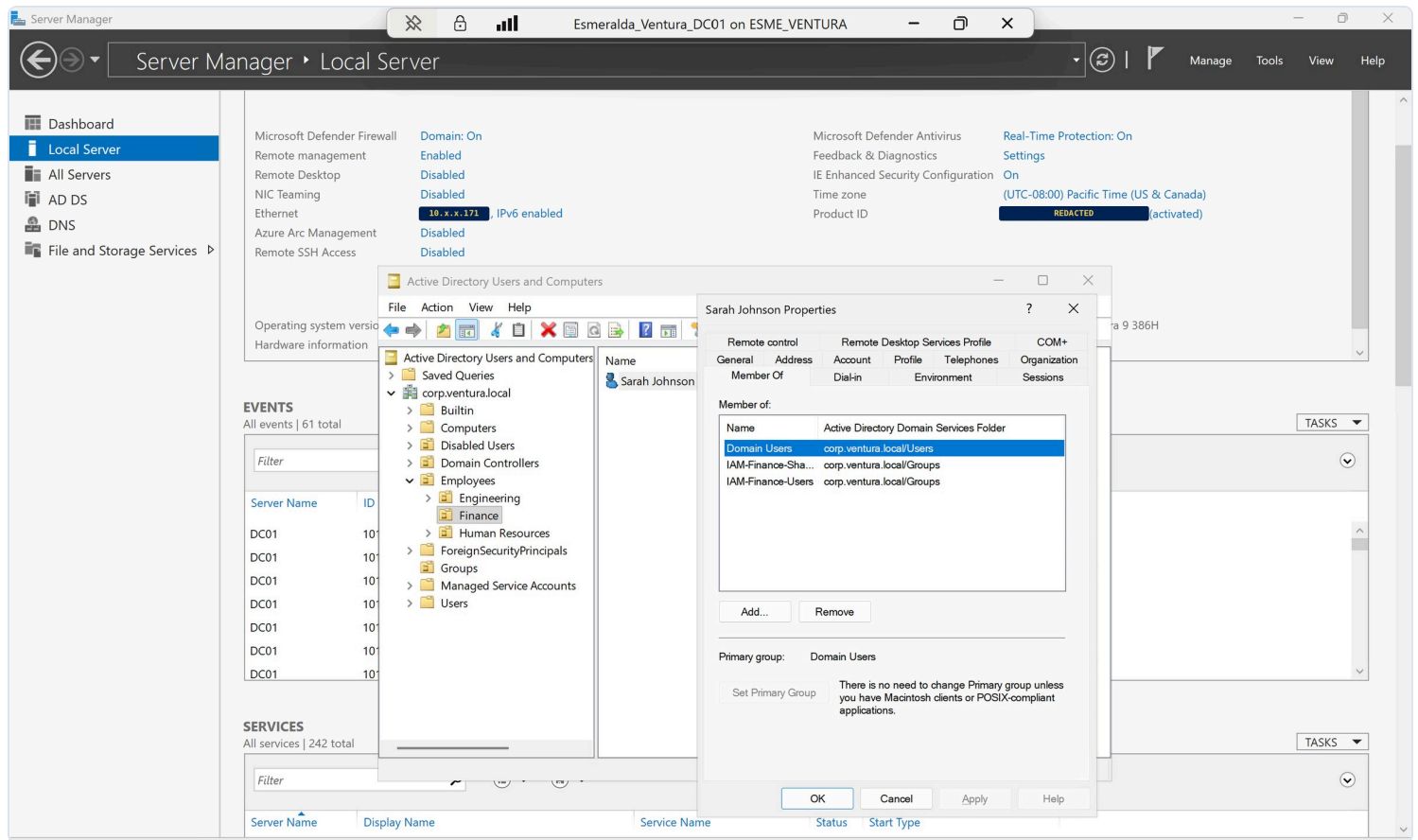


Figure 34. Member Of tab: Domain Users, IAM-Finance-SharedDrive and IAM-Finance-Users.

Expected versus observed

Expected	Observed	Result
Account created in Finance OU	DN in OU=Finance,OU=Employees	✓ Pass
Title, department, manager set	Financial Analyst, Finance, Jennifer Dovorany	✓ Pass
Baseline groups granted	IAM-Finance-SharedDrive , IAM-Finance-Users	✓ Pass
Enabled, audit folder created	Enabled True; C:\Lab\Audit present	✓ Pass

Role change: Financial Analyst to Cloud Engineer

HR record: same employee, still in Finance, new job title **Cloud Engineer**, manager unchanged, event **RoleChange**. Run with `Workday-RoleChange.csv`.

Reconciliation plan printed by the engine

System	Add	Remove	Keep
AD	none	none	IAM-Finance-SharedDrive, IAM-Finance-Users
Entra	IAM-Azure-Developers, IAM-Finance-M365, IAM-Intune-Corporate-Standard, IAM-Intune-Engineering	none	none
AVD	rg-avd Finance-Desktop	none	none

Entra and AVD still show pending additions because `-SkipCloud` prevented them from being applied. The engine keeps reporting this drift until a cloud run reconciles it.

Changes executed (1)

```
[SUCCESS] ActiveDirectory :: Update employee attributes :: sarah.johnson@corp.ventura.local
```

Verification output

Department / Title	Finance / Cloud Engineer
Manager	CN=Jennifer Dovorany,OU=Employees,DC=corp,DC=ventura,DC=local
DistinguishedName	CN=Sarah Johnson,OU=Finance,OU=Employees,DC=corp,DC=ventura,DC=local
Enabled	True
Groups	Domain Users, IAM-Finance-SharedDrive, IAM-Finance-Users

What this proves: idempotency. A naive script would remove and re-add every group on each run, adding log noise and brief access gaps. The engine saw the Finance groups were already correct and made no group writes. Only the attribute that changed was updated.

Expected	Observed	Result
Title updated, stays in Finance	Cloud Engineer; DN still in <code>OU=Finance</code>	✓ Pass
Manager unchanged	Jennifer Dovorany; no manager write issued	✓ Pass
Groups retained, no churn	Keep: <code>IAM-Finance-SharedDrive</code> , <code>IAM-Finance-Users</code>	✓ Pass
Account remains enabled	Enabled True	✓ Pass

Uncropped evidence follows: console output (Figure 35) and Active Directory Users and Computers (Figure 36).

```
Administrator: Windows Powe x + v Esmeralda_Ventura_DC01 on ESME_VENTURA
PS C:\Lab> .\Invoke-EmployeeIdentityLifecycle.ps1 -WorkdayCsv .\EmployeeLifeCycleTestCases\Workday-RoleChange.csv -ConfigPath .\AccessMappings.json -SkipCloud
Processing 10001 :: sarah.johnson@corp.ventura.local :: RoleChange

=====
ACCESS RECONCILIATION: sarah.johnson@corp.ventura.local
Event: RoleChange
Department: Finance
Role: Cloud Engineer
=====
AD
Add:
Remove:
Keep: IAM-Finance-SharedDrive, IAM-Finance-Users
Entra
Add: IAM-Azure-Developers, IAM-Finance-M365, IAM-Intune-Corporate-Standard, IAM-Intune-Engineering
Remove:
Keep:
AVD
Add: rg-avd|Finance-Desktop
Remove:
Keep:
[SUCCESS] ActiveDirectory :: Update employee attributes :: sarah.johnson@corp.ventura.local
PS C:\Lab> Get-ADUser sjohnson -Properties Department,Title,Manager | Select-Object Name,Department,Title,Manager,DistinguishedName,Enabled; Get-ADPrincipalGroupMem
bership sjohnson | Select-Object -ExpandProperty Name | Sort-Object

Name           : Sarah Johnson
Department      : Finance
Title           : Cloud Engineer
Manager         : CN=Jennifer Dovorany,OU=Employees,DC=corp,DC=ventura,DC=local
DistinguishedName : CN=Sarah Johnson,OU=Finance,OU=Employees,DC=corp,DC=ventura,DC=local
Enabled         : True

Domain Users
IAM-Finance-SharedDrive
IAM-Finance-Users

PS C:\Lab> |
```

Figure 35. RoleChange run: the AD section shows **Keep** for both Finance groups with nothing to add or remove. The only change executed is **Update employee attributes**. Verification shows Title Cloud Engineer.

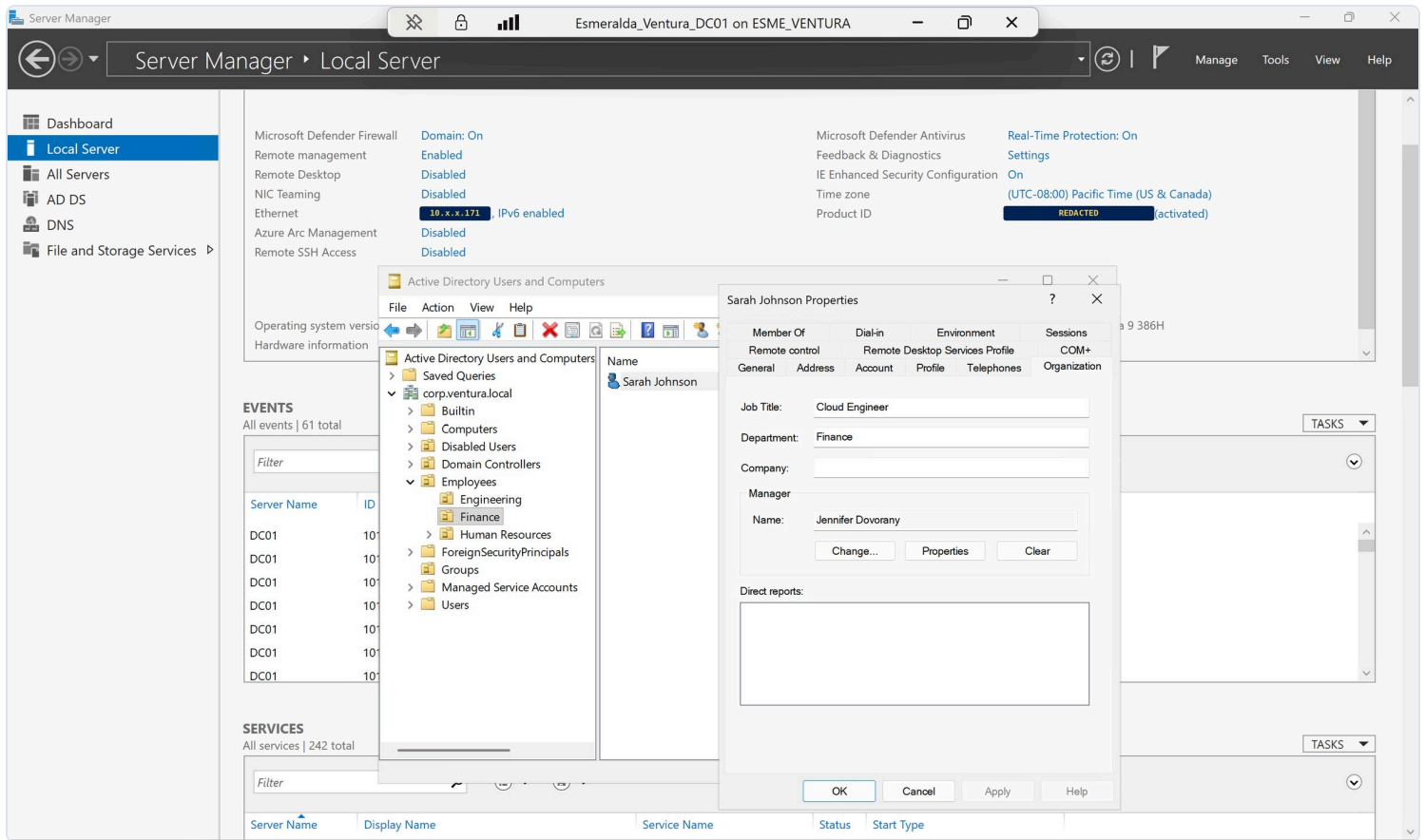


Figure 36. Organization tab after the role change: Job Title **Cloud Engineer**, Department **Finance**, Manager **Jennifer Dovorany**.

Department transfer: Finance to Engineering

HR record: same employee, now in **Engineering** as Cloud Engineer, new manager Robert Smith, event **DepartmentTransfer**. Run with `Workday-DepartmentTransfer.csv`.

Reconciliation plan printed by the engine

System	Add	Remove	Keep
AD	IAM-Engineering-SharedDrive , IAM-Engineering-Users	IAM-Finance-SharedDrive , IAM-Finance-Users	none
Entra	IAM-Azure-Developers , IAM-Engineering-M365 , IAM-Intune-Corporate-Standard , IAM-Intune-Engineering	none	none
AVD	rg-avd Engineering-Desktop	none	none

Changes executed (7)

```
[SUCCESS] ActiveDirectory :: Update employee attributes :: sarah.johnson@corp.ventura.local
[SUCCESS] ActiveDirectory :: Update manager :: sarah.johnson@corp.ventura.local
[SUCCESS] ActiveDirectory :: Move to OU 'OU=Engineering,OU=Employees,DC=corp,DC=ventura,DC=local'
[SUCCESS] ActiveDirectory :: Add group 'IAM-Engineering-SharedDrive' :: sarah.johnson@corp.ventura.local
[SUCCESS] ActiveDirectory :: Add group 'IAM-Engineering-Users' :: sarah.johnson@corp.ventura.local
[SUCCESS] ActiveDirectory :: Remove group 'IAM-Finance-SharedDrive' :: sarah.johnson@corp.ventura.local
[SUCCESS] ActiveDirectory :: Remove group 'IAM-Finance-Users' :: sarah.johnson@corp.ventura.local
```

Verification output

Department / Title	Engineering / Cloud Engineer
Manager	CN=Robert Smith,OU=Employees,DC=corp,DC=ventura,DC=local
DistinguishedName	CN=Sarah Johnson,OU=Engineering,OU=Employees,DC=corp,DC=ventura,DC=local
Enabled	True
Groups	Domain Users, IAM-Engineering-SharedDrive, IAM-Engineering-Users

Uncropped evidence follows: console output (Figure 37) and Active Directory Users and Computers (Figures 38–39).

```
Administrator: Windows Powe x + v Esmeralda_Ventura_DC01 on ESME_VENTURA
PS C:\Lab> .\Invoke-EmployeeIdentityLifecycle.ps1 -WorkdayCsv .\EmployeeLifeCycleTestCases\Workday-DepartmentTransfer.csv -ConfigPath .\AccessMappings.json -SkipClou
d

Processing 10001 :: sarah.johnson@corp.ventura.local :: DepartmentTransfer

=====
ACCESS RECONCILIATION: sarah.johnson@corp.ventura.local
Event: DepartmentTransfer
Department: Engineering
Role: Cloud Engineer
=====
AD
Add: IAM-Engineering-SharedDrive, IAM-Engineering-Users
Remove: IAM-Finance-SharedDrive, IAM-Finance-Users
Keep:
Entra
Add: IAM-Azure-Developers, IAM-Engineering-M365, IAM-Intune-Corporate-Standard, IAM-Intune-Engineering
Remove:
Keep:
AVD
Add: rg-avd|Engineering-Desktop
Remove:
Keep:
[SUCCESS] ActiveDirectory :: Update employee attributes :: sarah.johnson@corp.ventura.local
[SUCCESS] ActiveDirectory :: Update manager :: sarah.johnson@corp.ventura.local
[SUCCESS] ActiveDirectory :: Move to OU 'OU=Engineering,OU=Employees,DC=corp,DC=ventura,DC=local' :: sarah.johnson@corp.ventura.local
[SUCCESS] ActiveDirectory :: Add group 'IAM-Engineering-SharedDrive' :: sarah.johnson@corp.ventura.local
[SUCCESS] ActiveDirectory :: Add group 'IAM-Engineering-Users' :: sarah.johnson@corp.ventura.local
[SUCCESS] ActiveDirectory :: Remove group 'IAM-Finance-SharedDrive' :: sarah.johnson@corp.ventura.local
[SUCCESS] ActiveDirectory :: Remove group 'IAM-Finance-Users' :: sarah.johnson@corp.ventura.local
PS C:\Lab> Get-ADUser sjohnson -Properties Department,Title,Manager | Select-Object Name,Department,Title,Manager,DistinguishedName,Enabled; Get-ADPrincipalGroupMemb
ership sjohnson | Select-Object -ExpandProperty Name | Sort-Object

Name          : Sarah Johnson
Department    : Engineering
Title         : Cloud Engineer
Manager       : CN=Robert Smith,OU=Employees,DC=corp,DC=ventura,DC=local
DistinguishedName : CN=Sarah Johnson,OU=Engineering,OU=Employees,DC=corp,DC=ventura,DC=local
Enabled       : True

Domain Users
IAM-Engineering-SharedDrive
IAM-Engineering-Users

PS C:\Lab> |
```

Figure 37. DepartmentTransfer run: seven [SUCCESS] actions, including the move to the Engineering OU, two Engineering group additions and both Finance group removals. Verification shows the Engineering OU and manager Robert Smith.

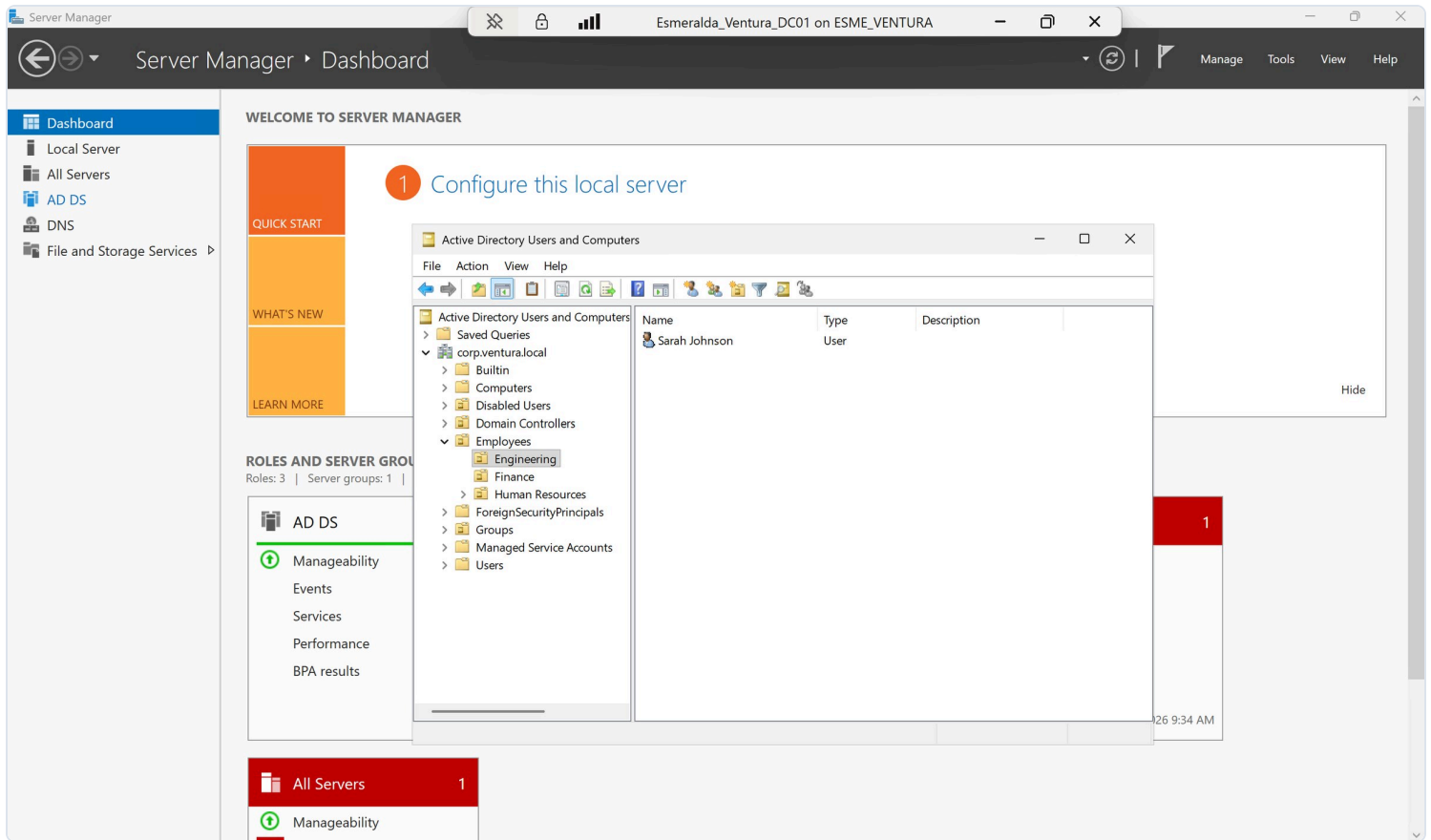


Figure 38. Active Directory Users and Computers: Sarah Johnson is now listed in the Engineering OU.

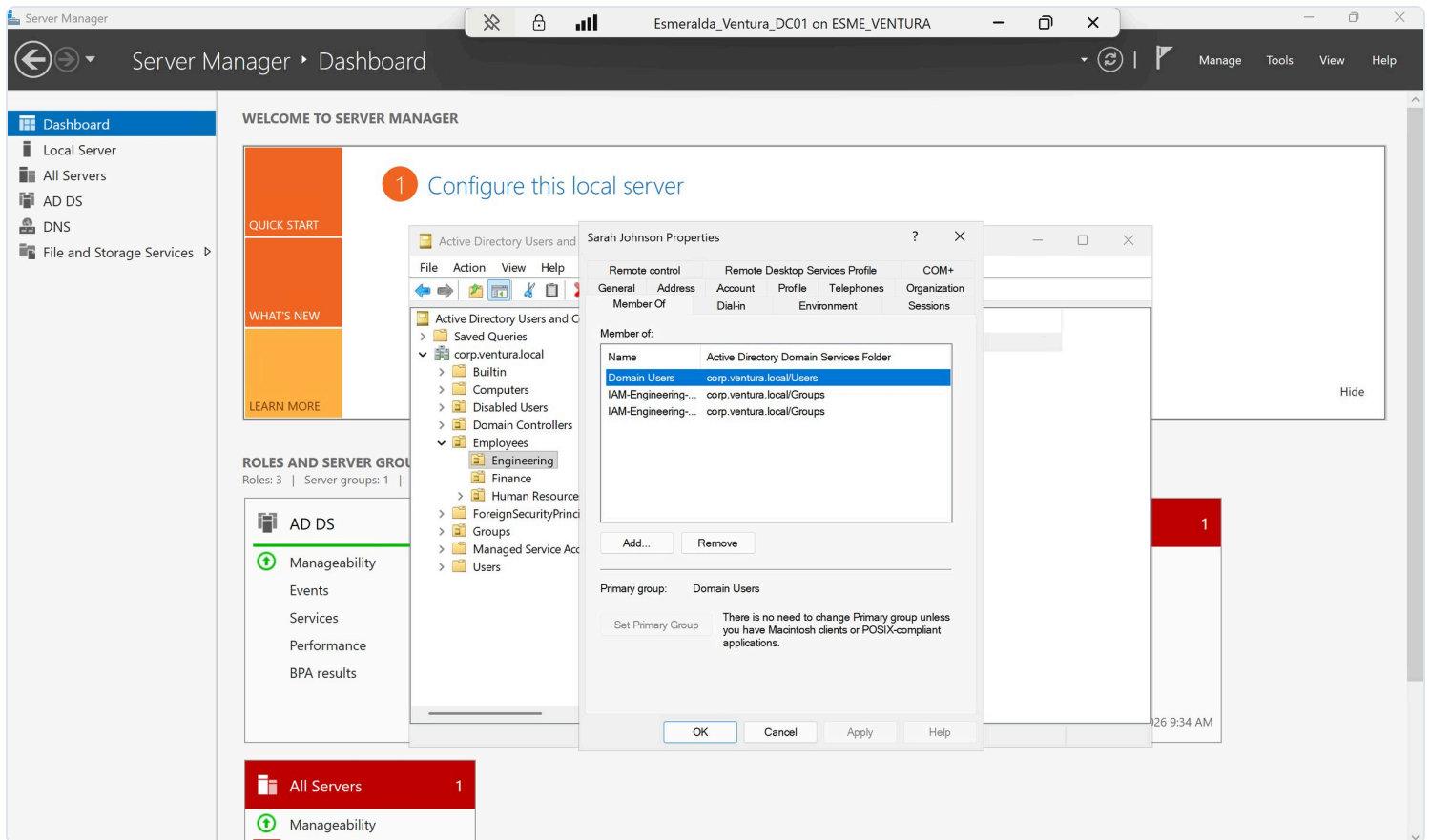


Figure 39. Member Of after the transfer: Domain Users, IAM-Engineering-SharedDrive and IAM-Engineering-Users. No Finance access remains.

Expected versus observed

Expected	Observed	Result
Moved to Engineering OU	DN in OU=Engineering,OU=Employees	✓ Pass
Department and manager updated	Engineering; Robert Smith	✓ Pass
Old access removed, new granted	-2 Finance, +2 Engineering in the same run	✓ Pass

Termination: secure offboarding

HR record: same employee, status **Terminated**, event **Termination**. Run with `Workday-Termination.csv`. For leavers the desired access is empty on every system.

Reconciliation plan printed by the engine

System	Add	Remove	Keep
AD	none	IAM-Engineering-SharedDrive, IAM-Engineering-Users	none
Entra	none	none	none
AVD	none	none	none

Changes executed (4)

```
[SUCCESS] ActiveDirectory :: Disable user :: sarah.johnson@corp.ventura.local
[SUCCESS] ActiveDirectory :: Remove group 'IAM-Engineering-Users' :: sarah.johnson@corp.ventura.local
[SUCCESS] ActiveDirectory :: Remove group 'IAM-Engineering-SharedDrive' :: sarah.johnson@corp.ventura.local
[SUCCESS] ActiveDirectory :: Move to OU 'OU=Disabled Users,DC=corp,DC=ventura,DC=local' :: sarah.johnson@corp.ventura.local
```

Verification output

Department / Title	Engineering / Cloud Engineer
DistinguishedName	CN=Sarah Johnson,OU=Disabled Users,DC=corp,DC=ventura,DC=local
Enabled	False
Groups	Domain Users only

Expected versus observed

Expected	Observed	Result
Account disabled	Enabled False	✓ Pass
All managed groups removed	Only Domain Users remains (unmanaged, untouched)	✓ Pass
Moved to quarantine OU	DN in OU=Disabled Users	✓ Pass

Why disable instead of delete. Deleting an account destroys its SID, group history and ownership links. Disabling and quarantining keeps evidence for audits, supports rehires, and gives the business time to hand over files. In the cloud path the same event also blocks Entra sign-in, revokes active sessions and removes AVD role assignments.

Uncropped evidence follows: console output (Figure 40) and Active Directory Users and Computers (Figure 41).

```
Administrator: Windows Powe x + v Esmeralda_Ventura_DC01 on ESME_VENTURA
Domain Users
IAM-Engineering-SharedDrive
IAM-Engineering-Users

PS C:\Lab>
> .\Invoke-EmployeeIdentityLifecycle.ps1 -WorkdayCsv .\EmployeeLifeCycleTestCases\Workday-Termination.csv -ConfigPath .\AccessMappings.json -SkipCloud

Processing 10001 :: sarah.johnson@corp.ventura.local :: Termination

=====
ACCESS RECONCILIATION: sarah.johnson@corp.ventura.local
Event: Termination
Department: Engineering
Role: Cloud Engineer
=====
AD
Add:
Remove: IAM-Engineering-SharedDrive, IAM-Engineering-Users
Keep:
Entra
Add:
Remove:
Keep:
AVD
Add:
Remove:
Keep:
[SUCCESS] ActiveDirectory :: Disable user :: sarah.johnson@corp.ventura.local
[SUCCESS] ActiveDirectory :: Remove group 'IAM-Engineering-Users' :: sarah.johnson@corp.ventura.local
[SUCCESS] ActiveDirectory :: Remove group 'IAM-Engineering-SharedDrive' :: sarah.johnson@corp.ventura.local
[SUCCESS] ActiveDirectory :: Move to OU 'OU=Disabled Users,DC=corp,DC=ventura,DC=local' :: sarah.johnson@corp.ventura.local
PS C:\Lab> Get-ADUser sjohnson -Properties Department,Title,Manager | Select-Object Name,Department,Title,Manager,DistinguishedName,Enabled; Get-ADPrincipalGroupMem
bership sjohnson | Select-Object -ExpandProperty Name | Sort-Object

Name
: Sarah Johnson
Department
: Engineering
Title
: Cloud Engineer
Manager
: CN=Robert Smith,OU=Employees,DC=corp,DC=ventura,DC=local
DistinguishedName
: CN=Sarah Johnson,OU=Disabled Users,DC=corp,DC=ventura,DC=local
Enabled
: False

Domain Users

PS C:\Lab> |
```

Figure 40. Termination run: Disable user, removal of both Engineering groups, and Move to OU Disabled Users. Verification shows **Enabled: False** and only Domain Users remaining.

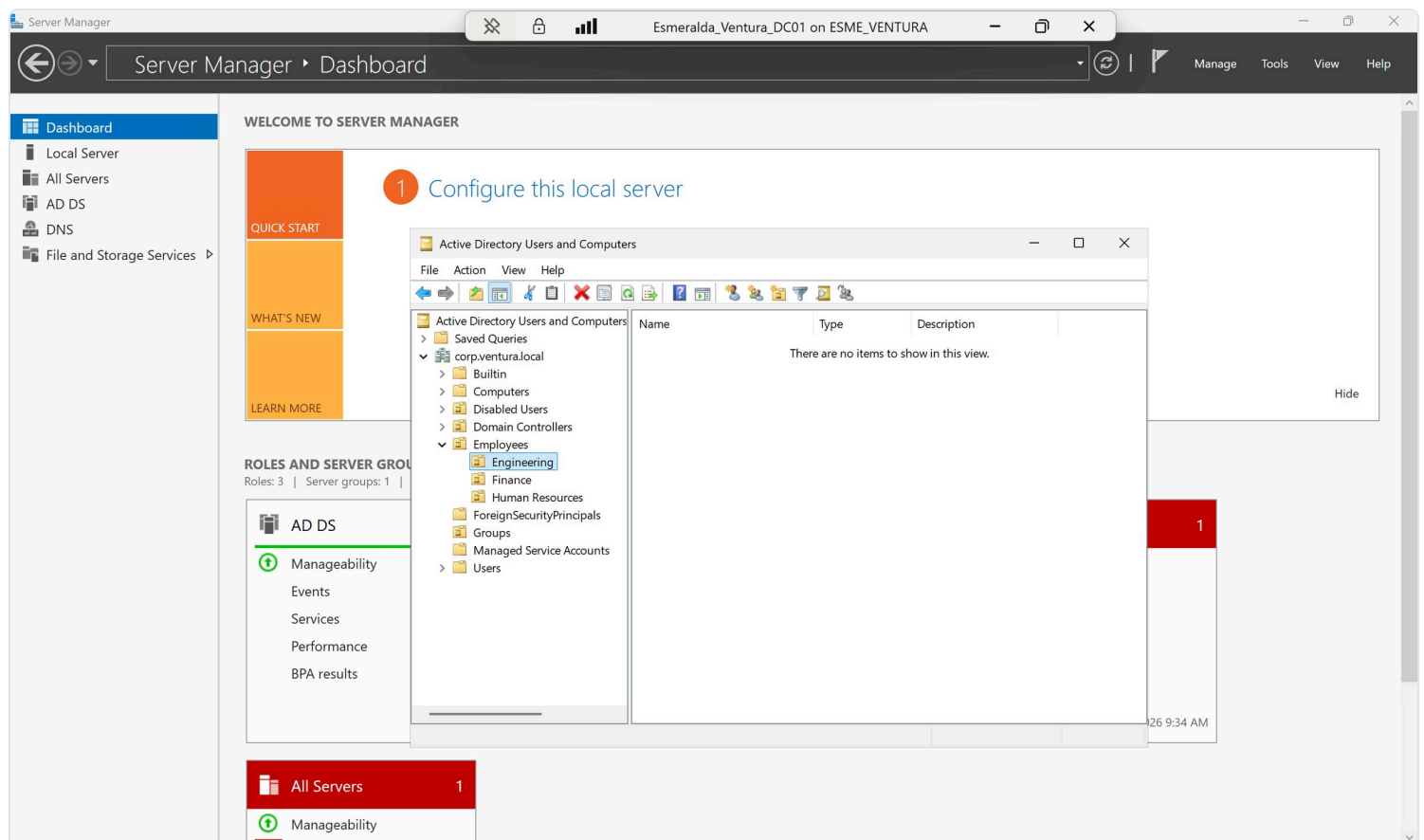


Figure 41. Active Directory Users and Computers after termination: the **Engineering** OU is empty. The account now lives only in Disabled Users.

06 Results scorecard

14/14 Expected outcomes verified across the four tests	17 Directory changes, all logged with a Success result	0 Unmanaged groups touched (Domain Users intact throughout)	0 Redundant writes when access was already correct
--	--	---	--

Changes applied per lifecycle event, by action type

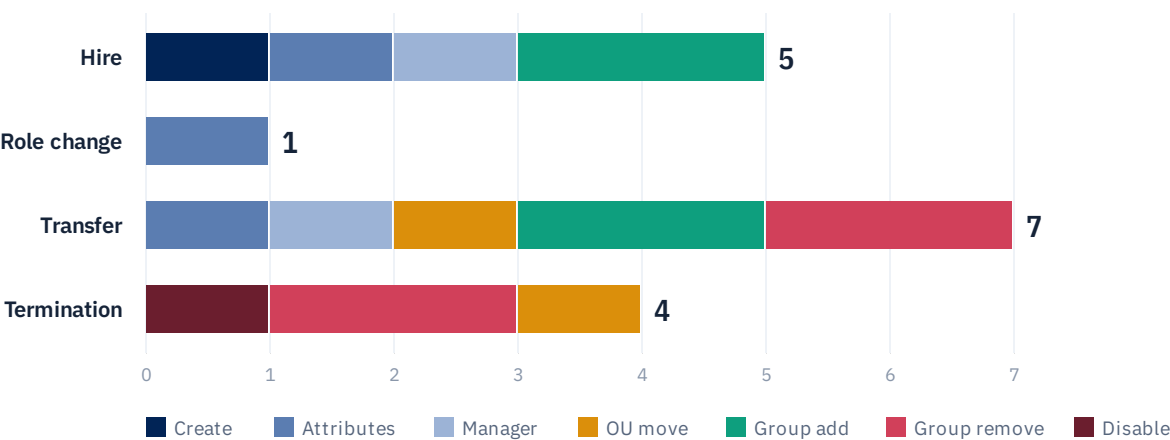


Figure 42. All 17 changes, taken from the [SUCCESS] lines in the console output of each test.

Manual process versus the engine

Event	Manual process in ADUC	With the engine
Hire	New User wizard, then Properties for title, department and manager, then Member Of for each group. Group choice often copied from a peer.	One command. Groups are computed from policy, the manager is resolved from the HR record, and the password must change at first sign-in.
RoleChange	Edit title. Decide by hand which entitlements the new role needs and which to drop.	Role overlay recalculated. Correct access is kept and only real differences are applied.
Transfer	Move OU, change manager, add new groups. Old groups are frequently left in place.	Old department groups removed in the same run the new ones are granted, so access can't pile up.
Termination	Depends on a ticket arriving and a checklist being followed across AD, cloud and VDI consoles.	Disable, strip managed access, quarantine and (in the cloud) revoke sessions, all from a single HR status change.
Evidence	Ticket notes, if any.	Structured JSONL audit record for every action, including failures.

07 Security and governance

A lab is allowed shortcuts. A production identity system isn't. This section separates what is already enforced from what I would harden before running against a real company directory.

Controls already implemented

Change safety

-WhatIf, confirmation prompts at high impact, and a plan-only - AuditOnly mode. Strict mode and stop-on-error.

Least impact

Prefix-scoped group management, matching on employee ID, and per-employee error isolation with failures logged.

Account hygiene

New users must change their password at first login. Leavers are disabled and quarantined. OUs are protected from deletion. Cloud sessions are revoked on exit.

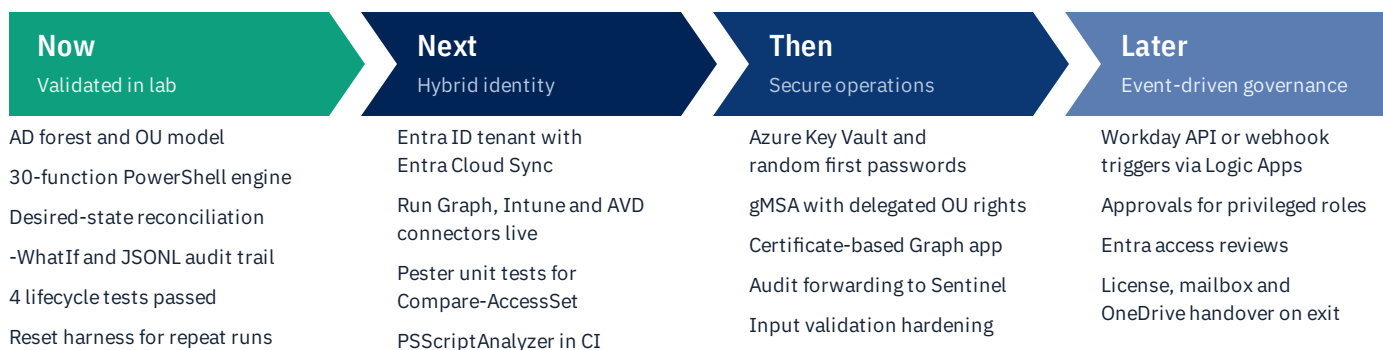
Lab shortcuts and their production replacements

Lab shortcut	Risk	Production hardening
Initial password stored in plaintext in <code>AccessMappings.json</code>	Anyone with file access learns every new hire's first password	Generate a unique random password per user at run time, or use an Entra Temporary Access Pass. Keep secrets in Azure Key Vault through PowerShell SecretManagement.
Runs as the domain Administrator	A bug or compromise has full domain control	Dedicated gMSA with delegated rights only on the Employees and Disabled Users OUs and the <code>IAM-</code> groups.
Interactive Graph sign-in with <code>Directory.ReadWrite.All</code>	Over-privileged, and can't run unattended	App registration with certificate authentication, scoped to <code>User.ReadWrite.All</code> and <code>GroupMember.ReadWrite.All</code> .
CSV values placed into AD filter strings	Malformed input could change query meaning	Validate <code>EmployeeId</code> as digits and UPNs against a pattern, and escape quotes before building filters.
Audit log on the local disk of the DC	Can be altered or lost with the server	Forward to Microsoft Sentinel or another SIEM with immutable retention, and alert on Failed results.
Manually triggered with a CSV file	Depends on an operator running it on time	Scheduled Azure Automation runbook on a Hybrid Worker, fed by the Workday API, plus a nightly Reconcile run to catch drift.

How this report was sanitized

The following were removed from the screenshots before publication: plaintext lab passwords, MAC addresses and the Hyper-V MAC range, IPv6 and DHCPv6 identifiers, the domain SID and object GUID, the Windows product ID, local user paths, and personal file names. Password fields were blurred even though they were masked, because masking still reveals length. Internal IPv4 addresses are shown only as `10.x.x.171` so the network design stays readable. Screenshots showing remote-access overlay network details were left out entirely. No screenshot was cropped: every frame is shown in full, so the VM name `Esmeralda_Ventura_DC01`, host name `ESME_VENTURA` and domain `corp.ventura.local` remain visible as proof of authorship.

08 Roadmap and engineering review



Each stage builds on the same engine. New capabilities arrive as connectors or policy entries, not rewrites.

Figure 43. Delivery roadmap from lab-validated AD automation to fully governed hybrid identity.

Engineering review: what I would refine next

Reviewing my own test output critically surfaced improvements that are small in code but meaningful in operations.

Observation from the tests	Refinement
"Update employee attributes" runs on every event, even when title and department are unchanged.	Compare attribute values before writing, so the audit log only records real changes and stays high-signal.
A terminated account keeps its manager and title.	Clear the manager link, stamp a termination date in the description, and hide the account from address lists.
The Cloud Engineer role currently adds cloud entitlements only.	Introduce role-scoped AD groups (<code>IAM-CloudEngineering-Users</code> and <code>-SharedDrive</code>). This is already designed in the project README and only needs a policy entry and two groups.
Cloud plans are printed but not applied under <code>-SkipCloud</code> .	Connect a developer tenant and repeat the four-test campaign with cloud verification in Entra admin center and Intune.
Tests are verified by reading output.	Codify expected results as Pester assertions so the full campaign becomes one automated regression suite.

09 Skills demonstrated

Every capability below is backed by evidence in this report, not just listed. Page references point to the screenshots, code or results that prove it.

Identity and access management

Joiner-Mover-Leaver design, RBAC with department baselines and role overlays, least privilege, and privilege-creep prevention.

Evidence: pages 3, 17, 21, 26

Active Directory engineering

New forest at Windows Server 2025 level, DC promotion, FSMO and global catalog verification, OU and security group design.

Evidence: pages 12, 17; Figures 12–24

PowerShell automation

Advanced functions, `CmdletBinding` and `ShouldProcess`, parameter validation, strict mode, structured error handling, JSON and CSV processing.

Evidence: pages 5, 20, 21, 22

Microsoft cloud identity

Microsoft Graph SDK for Entra ID groups, sign-in blocking and session revocation. Intune group targeting. AVD RBAC with Az modules.

Evidence: pages 4, 17, 20

Virtualization and networking

Hyper-V Generation 2 VMs, external virtual switching, resource sizing, static addressing, DNS configuration and connectivity testing.

Evidence: pages 6, 7; Figures 4–9

Troubleshooting

Evidence-based diagnosis of a misleading static-IP validation warning, with a documented and reversible decision.

Evidence: page 7; Figures 10–11

Testing and quality

Scenario-based test plan, expected-versus-observed verification from two independent sources, and a reset harness for repeatable runs.

Evidence: pages 26, 27, 30, 32, 35, 37

Security mindset

Audit trails by design, blast-radius limiting, threat-aware hardening plan, and careful redaction of secrets and identifiers.

Evidence: pages 5, 38

Technical communication

Architecture diagrams, clear trade-off reasoning, honest scoping of what was and wasn't validated, and a self-review with next steps.

Evidence: throughout; page 39

Access that follows people automatically, correctly, and with proof.

This project shows how I approach infrastructure work. I build the environment myself, design for safety before speed, verify results independently, and document decisions so others can trust and extend them.

Engineer

Esmeralda Cabrera Ventura

Stack

Windows Server 2025, AD DS, PowerShell, Graph

Status

AD path validated